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JAMB Biology: The Complete Course

Concepts, worked examples and exercises for the UTME

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Chapter 1: Living Things and Their Classification

Biology is the scientific study of living things. It asks what life is, how living organisms are built and how they work, how they reproduce and change through time, and how they depend on one another and on their surroundings. A farmer in Ekiti deciding when to plant yam, a nurse in Kano explaining malaria prevention and a researcher in Ibadan breeding a better variety of cassava are all using biological knowledge, whether or not they call it by that name.

This opening chapter lays the foundation for everything that follows. We begin with the characteristics that distinguish living things from non-living things, then examine the levels at which living matter is organised, and finally study the system scientists use to name and group the millions of kinds of organisms on Earth. The classification of organisms is a favourite topic in the UTME, and candidates are expected to recognise the major groups, to know their distinguishing features and to give correct examples.

1.1 The branches of biology and the scientific method

Because the living world is so vast, biology is divided into branches. **Botany** is the study of plants and **zoology** is the study of animals. **Microbiology** deals with organisms too small to be seen without a microscope, such as bacteria, many fungi and protozoa. **Cytology** is the study of cells, **genetics** the study of heredity and variation, **ecology** the study of the relationships between organisms and their environment, and **physiology** the study of how the parts

of an organism function. **Anatomy** describes structure, **morphology** describes external form, **taxonomy** is concerned with naming and classifying, and **entomology** is the study of insects. **Parasitology** focuses on parasites, **mycology** on fungi, and **pathology** on diseases and their causes.

Biologists generally work by the **scientific method**. The process begins with an observation that raises a question, for example why the leaves of cocoa seedlings kept in a dark shed are pale. The scientist then proposes a **hypothesis**, a testable explanation, such as the suggestion that light is required for the formation of chlorophyll. An **experiment** is designed to test the hypothesis. In a good experiment only one factor, the **independent variable**, is changed deliberately; the **dependent variable** is measured, and all other conditions are kept constant. A **control** is set up in which the factor under investigation is left unchanged so that results can be compared fairly. Results are recorded, analysed and used to draw a conclusion, and the experiment is repeated to confirm that the result is reliable. If many experiments support a hypothesis over a long period, it may grow into a **theory**.

> **Key point.** A hypothesis must be testable and, in principle, capable of being shown to be false. A control experiment is not a repeat of the main experiment; it differs from it in exactly one respect, the factor being tested.

1.2 Characteristics of living things

There is no single property that defines life, but all living organisms show a set of characteristics, and non-living objects do not show all of them. Seven are normally listed, and it is useful to remember them by the initials of the

words *movement, respiration, sensitivity, growth, reproduction, excretion and nutrition.*

Movement. Living things change position or move parts of their bodies. Animals usually move the whole body from place to place, which is called **locomotion**. Most plants are rooted and cannot move from place to place, but they show movement of parts: shoots bend towards light, the leaflets of the sensitive plant *Mimosa pudica* fold when touched, and flowers open and close. Some unicellular organisms such as *Euglena* swim by means of a flagellum.

Respiration. All living cells release energy from food. This chemical process, which usually uses oxygen to break down glucose, takes place continuously and is distinct from breathing, which is merely the physical movement of air in and out of the lungs. Energy released in respiration drives every other activity of life.

Sensitivity (irritability). Organisms detect changes in their surroundings, called **stimuli**, and respond to them. A goat withdraws from a thorn, a lizard basks in the morning sun, and the roots of a germinating maize seed grow downwards in response to gravity.

Growth. Growth is a permanent increase in size and dry mass, brought about by cell division and cell enlargement. It is not simply swelling: a dry bean that absorbs water increases in size but not in dry mass, so it has not grown.

Reproduction. Living things produce new individuals of their own kind, so that the species continues even though individuals die. Reproduction may be asexual, involving one parent, or sexual, usually involving two.

Excretion. Metabolism produces waste products such as carbon dioxide, water and urea, some of which are harmful if allowed to accumulate. Excretion is the removal of the waste products of metabolism from the body. It

must be distinguished from **egestion**, the removal of undigested food, which has never been part of the body's metabolism.

Nutrition. Organisms take in or manufacture the materials they need for energy, growth and repair. Green plants make their own food by photosynthesis; animals obtain ready-made food by feeding.

Some textbooks add further characteristics, such as **adaptation** to the environment and **death**. A non-living thing may show one of the seven features. A candle flame, for example, uses oxygen, produces carbon dioxide and appears to grow, and a car moves and uses fuel. Neither is alive, because neither shows all the features together and neither carries them out by controlled chemical processes within cells.

Differences between plants and animals

Feature	Plants	Animals
Nutrition	Autotrophic, making food by photosynthesis	Heterotrophic, depending on ready-made food
Cell wall	Present, made of cellulose	Absent
Chlorophyll	Present in chloroplasts	Absent
Locomotion	Usually fixed in one place	Usually able to move from place to place
Growth	Continues throughout life at growing points	Generally stops at maturity, occurring throughout the body
Food reserve	Starch, oils	Glycogen, fat
Response to stimuli	Slow	Quick, through a nervous system

These differences are generalisations. The pitcher plant traps and digests insects, and *Euglena* carries chloroplasts yet swims freely, which shows why classification needs more than one criterion.

1.3 Levels of organisation

Living matter is organised in a hierarchy. In a multicellular organism, cells of the same kind that perform one function are grouped into a **tissue**. Examples are muscle tissue, which contracts, and xylem tissue, which conducts water in plants. Different tissues combine to form an **organ**, such as the stomach, the heart or the leaf. Organs working together to perform a major function form an **organ system**, for example the digestive, circulatory, excretory and nervous systems. All the systems together make up the **organism**.

Beyond the individual, we speak of a **population**, all the individuals of one species living in an area, a **community**, all the populations living and interacting in that area, an **ecosystem**, the community together with its non-living environment, and the **biosphere**, the part of the Earth where life exists. Thus the sequence from the smallest unit to the largest is: cell, tissue, organ, organ system, organism, population, community, ecosystem, biosphere.

Organisms also differ in how many cells they contain. **Unicellular** organisms, such as *Amoeba*, *Paramecium* and *Chlamydomonas*, consist of a single cell that performs all life functions. **Multicellular** organisms contain many cells, with division of labour among them. A **colonial** organism such as *Volvox* is a hollow ball of cells in which the cells cooperate but show little specialisation.

1.4 The need for classification

More than a million and a half species have been named and described, and many more remain unknown. Without an orderly system, studying them would be impossible. **Classification**, or taxonomy, is the arrangement of organisms into groups on the basis of their similarities and differences, ideally reflecting their evolutionary relationships. A good system gives every organism a unique and universally understood name, makes identification easier, shows relationships, and allows biologists in different countries to be certain that they are discussing the same organism. Local names are unreliable: the same plant may carry different names in Yoruba, Igbo and Hausa, while one name may be used for different plants in different towns.

The taxonomic hierarchy

Organisms are placed in a series of groups, each contained within a larger one. The groups, called **taxa** (singular **taxon**), run from the largest to the smallest: **Kingdom, Phylum (Division in plants), Class, Order, Family, Genus, Species**. A convenient memory aid is the sentence "King Philip Came Over For Good Soup". The basic unit is the **species**, a group of organisms that resemble one another closely and can interbreed to produce fertile offspring. A horse and a donkey can mate, but their offspring, the mule, is sterile, so they are separate species.

The following table shows how two familiar organisms are classified.

Taxon	Man	Maize
Kingdom	Animalia	Plantae
Phylum / Division	Chordata	Angiospermophyta
Class	Mammalia	Monocotyledonae
Order	Primates	Poales
Family	Hominidae	Poaceae
Genus	*Homo*	*Zea*
Species	*Homo sapiens*	*Zea mays*

Binomial nomenclature

The system of naming used worldwide was devised by the Swedish naturalist **Carolus Linnaeus** in the eighteenth century. In **binomial nomenclature**, each organism is given a two-part Latin name. The first part is the **genus**, written with a capital initial, and the second is the **specific epithet**, written in lower case. The whole name is printed in italics, or underlined when handwritten, as in *Homo sapiens*, *Mangifera indica* (mango), *Elaeis guineensis* (oil palm) and *Theobroma cacao* (cocoa). Closely related species share a genus: the domestic dog *Canis familiaris* and the wolf *Canis lupus* both belong to *Canis*.

> **Key point.** In writing a scientific name, the genus begins with a capital letter and the species name with a small letter, and both are italicised or underlined separately. Writing *homo Sapiens* or *Homo Sapiens* is wrong.

1.5 The five kingdoms

The most widely used scheme in secondary schools divides living things into five kingdoms, namely Monera, Protista, Fungi, Plantae and Animalia. The basis is the type of cell, the number of cells, the method of nutrition and the complexity of the body.

Kingdom Monera

Monera includes the bacteria and the cyanobacteria (blue-green algae). They are **prokaryotes**: their cells have no true nucleus enclosed by a membrane and no membrane-bound organelles such as mitochondria. The genetic material lies free in the cytoplasm. They are microscopic, mostly unicellular, and have cell walls made of a substance different from cellulose. Some are autotrophic, others are saprophytic or parasitic. Examples are *Escherichia coli*, *Salmonella typhi* (the cause of typhoid) and *Nostoc*.

Kingdom Protista

Protista comprises mostly unicellular **eukaryotes**, organisms whose cells have a true nucleus and membrane-bound organelles. It includes the protozoa such as *Amoeba*, *Paramecium* and *Plasmodium* (the malaria parasite), and many algae such as *Euglena* and *Chlamydomonas*. Some are animal-like, feeding on food, while others are plant-like, with chloroplasts.

Kingdom Fungi

Fungi are eukaryotic organisms without chlorophyll. They feed by absorbing dissolved food from dead organic matter (saprophytes) or from living hosts (parasites). Their bodies are usually made of thread-like **hyphae** that together form a **mycelium**, and their cell walls contain **chitin**, not cellulose. They

reproduce by spores and store food as glycogen. Examples are *Mucor* (pin mould), *Rhizopus* (bread mould), *Saccharomyces* (yeast), *Agaricus* (mushroom) and *Penicillium*.

Kingdom Plantae

Plantae contains multicellular eukaryotes with cellulose cell walls and chloroplasts, which make food by photosynthesis. The kingdom is divided into the following divisions.

- **Bryophyta** (mosses and liverworts): small plants without true roots, stems or leaves, lacking vascular tissue, found in damp places, and reproducing by spores produced in a capsule. Example: *Funaria*.
- **Pteridophyta** (ferns): plants with true roots, stems and leaves, and with vascular tissue (xylem and phloem). They reproduce by spores, usually carried in sporangia on the underside of leaves, and have no seeds. Example: *Dryopteris*, the male fern.
- **Spermatophyta** (seed plants): plants that reproduce by seeds. They are divided into the **Gymnospermae**, with naked seeds not enclosed in an ovary, usually borne in cones, as in pine; and the **Angiospermae**, the flowering plants, whose seeds are enclosed in a fruit developing from an ovary.

The angiosperms are split into two classes. **Monocotyledons** have one seed leaf, parallel-veined leaves, fibrous roots, scattered vascular bundles in the stem and flower parts in threes; examples are maize, rice, sugarcane, banana and grasses. **Dicotyledons** have two seed leaves, net-veined leaves, a tap root, vascular bundles arranged in a ring and flower parts in fours or fives; examples are beans, groundnut, mango, hibiscus and orange.

Feature	Monocotyledon	Dicotyledon
Cotyledons in seed	One	Two
Leaf venation	Parallel	Reticulate (net-like)
Root system	Fibrous	Tap root with laterals
Vascular bundles	Scattered	In a ring
Cambium	Absent, so little secondary growth	Present, so secondary thickening occurs
Flower parts	Multiples of three	Multiples of four or five

Kingdom Animalia

Animalia comprises multicellular eukaryotes whose cells lack walls and chloroplasts and which feed heterotrophically, usually by ingestion. Animals are divided into phyla, of which the following are the most important.

- **Porifera** (sponges): aquatic animals with bodies full of pores, with no true tissues. Example: *Sycon*.
- **Cnidaria** (coelenterates): aquatic animals with radial symmetry, a body wall of two layers, a single opening serving as mouth and anus, and stinging cells. Examples: *Hydra*, jellyfish, sea anemone.
- **Platyhelminthes** (flatworms): flattened, unsegmented, bilaterally symmetrical worms with no body cavity. Examples: *Taenia solium* (tapeworm), *Schistosoma* (blood fluke), and the planarian.
- **Nematoda** (roundworms): cylindrical, unsegmented worms with a complete gut. Examples: *Ascaris lumbricoides*, hookworm, and *Wuchereria bancrofti*, which causes elephantiasis.
- **Annelida** (segmented worms): bodies divided into ring-like segments, with bristles (chaetae) in many. Examples: earthworm, leech.

- **Mollusca**: soft-bodied animals, usually with a shell, a muscular foot and a mantle. Examples: snail, mussel, oyster, octopus.
- **Arthropoda**: the largest phylum, with jointed legs, an external skeleton of chitin (exoskeleton) and segmented bodies. It is divided into classes.
- **Chordata**: animals with a notochord, a dorsal hollow nerve cord and gill slits at some stage of life. The vertebrates, with a backbone, belong here.

Classes of Arthropoda

Class	Body regions	Legs	Antennae	Examples
Insecta	Head, thorax, abdomen	3 pairs	1 pair	Housefly, grasshopper, mosquito, bee
Arachnida	Cephalothorax, abdomen	4 pairs	None	Spider, scorpion, tick
Crustacea	Cephalothorax, abdomen	5 or more pairs	2 pairs	Crab, prawn, lobster
Chilopoda	Head and trunk	1 pair per segment	1 pair	Centipede
Diplopoda	Head and trunk	2 pairs per segment	1 pair	Millipede

Classes of Chordata (vertebrates)

Class	Skin covering	Breathing	Body temperature	Reproduction	Examples
Pisces (fishes)	Scales	Gills	Cold-blooded	External fertilisation, eggs in water	Tilapia, catfish
Amphibia	Moist, smooth, no scales	Gills as larva; lungs and skin as adult	Cold-blooded	Eggs laid in water	Toad, frog
Reptilia	Dry scales	Lungs	Cold-blooded	Leathery-shelled eggs on land	Lizard, snake, crocodile
Aves (birds)	Feathers	Lungs with air sacs	Warm-blooded	Hard-shelled eggs	Fowl, eagle
Mammalia	Hair or fur	Lungs	Warm-blooded	Young usually born alive and suckled	Goat, bat, man, whale

Mammals possess mammary glands, a diaphragm, external ears (pinnae) in most, and four-chambered hearts. Birds and mammals have a four-chambered heart; amphibians have three chambers; most reptiles have three chambers with a partial division of the ventricle; fishes have two.

Key point. The bat flies like a bird, but it has hair, mammary glands and gives birth to live young, so it is a mammal. The whale lives in the sea like a fish, but it breathes with lungs and suckles its young, so it is also a mammal. Always classify by fundamental structural features, not by habitat or mode of movement.

1.6 Viruses and the borderline of life

Viruses do not fit comfortably into any of the five kingdoms. They are extremely small, usually between 20 and 300 nanometres, and can be seen only with an electron microscope. A virus has no cytoplasm and no organelles; it consists of a core of nucleic acid, either DNA or RNA but never both, surrounded by a protein coat called a **capsid**. Outside a living cell, a virus shows no sign of life and can even be crystallised. Inside a host cell, however, it takes over the cell's machinery and makes copies of itself. Viruses are therefore **obligate parasites**. Examples include the viruses that cause polio, measles, influenza, rabies and AIDS, the mosaic virus of cassava and the bacteriophage that infects bacteria. Because they reproduce only in hosts and do not grow, respire or respond independently, many biologists regard them as non-living entities on the borderline of life.

1.7 Using keys to identify organisms

A **dichotomous key** is a tool for identifying organisms, built from numbered pairs of contrasting statements. At each step the user chooses the statement that fits the specimen and is directed either to another pair or to a name. For example, to identify four animals:

1. (a) Body with a backbone: go to 2. (b) Body without a backbone: go to 3.
2. (a) Body covered with feathers: bird. (b) Body covered with scales: reptile.
3. (a) Body with three pairs of legs: insect. (b) Body with four pairs of legs: spider.

Good keys use clear observable features, not size or colour, which vary with age, and each statement must be exclusive of its partner.

1.8 Worked examples

WORKED EXAMPLE 1.1

A specimen has a body divided into head, thorax and abdomen, three pairs of jointed legs and one pair of antennae. To which class of which phylum does it belong? Give a reason for your answer.

Solution: Jointed legs and a segmented body covered with a chitinous exoskeleton place the animal in the phylum Arthropoda. Within the arthropods, three body regions, three pairs of legs and a single pair of antennae are the features of the class Insecta. A spider would have two body regions, four pairs of legs and no antennae; a crab would have two pairs of antennae and five or more pairs of legs. The specimen is therefore an insect in the phylum Arthropoda.

WORKED EXAMPLE 1.2

Name the kingdom of each organism and give one reason: (a) *Rhizopus* growing on stale bread; (b) *Escherichia coli*; (c) a mango tree; (d) *Paramecium*.

Solution: (a) *Rhizopus* is in Kingdom Fungi, because it has hyphae, chitin in its walls and no chlorophyll, feeding saprophytically. (b) *E. coli* is in Kingdom Monera, because it is a bacterium with a prokaryotic cell and no true nucleus. (c) The mango tree is in Kingdom Plantae, because it is multicellular, has cellulose walls and makes food by photosynthesis. (d) *Paramecium* is in Kingdom Protista, because it is a unicellular eukaryote with a nucleus.

WORKED EXAMPLE 1.3

A student places 20 dry cowpea seeds in a beaker of water overnight and finds that the seeds are bigger and heavier in the morning. She claims the seeds have grown. Evaluate her claim.

Solution: Growth is a permanent increase in dry mass resulting from cell division and enlargement, with synthesis of new materials. The seeds became larger and heavier only because they absorbed water by imbibition. If the seeds were dried in an oven and weighed again, their dry mass would be the same as, or slightly less than, the original. So no growth has occurred. If the seeds were left to germinate and the dry mass of the seedlings measured after some days, a true increase might still be absent at first, since the seedling uses stored food in respiration; real growth in dry mass begins only when the seedling has leaves that photosynthesise.

WORKED EXAMPLE 1.4

Write the correct scientific names of the oil palm (genus Elaeis, species guineensis) and of the domestic cat (genus Felis, species catus), and explain why *Felis catus* and *Felis silvestris* are separate species although they are in the same genus.

Solution: The names are *Elaeis guineensis* and *Felis catus*, with the genus capitalised, the species in lower case, and both italicised. Species within a genus share many features, which is why they are placed together, but they are distinct breeding units. Members of different species either do not interbreed in nature or, if they do, produce infertile or non-viable offspring. This reproductive isolation is why *Felis catus* and *Felis silvestris* are recorded as two species.

WORKED EXAMPLE 1.5

Arrange in correct order from the largest to the smallest taxon: genus, class, family, kingdom, species, phylum, order. Then state which two taxa are shared by the lion (*Panthera leo*) and the leopard (*Panthera pardus*) from among genus and species.

Solution: The order is kingdom, phylum, class, order, family, genus, species. The lion and the leopard have the same first word, *Panthera*, so they share the genus (and every taxon above it), but their second words differ, so they are different species. The only one of the two named taxa that they share is the genus.

1.9 Exercise

Exercise 1.1

1. Which of the following is the correct order of organisation in a multicellular organism: tissue, cell, organ system, organ?
2. State two ways in which excretion differs from egestion.
3. Which of the following is NOT a characteristic of all living things: (a) respiration, (b) locomotion from place to place, (c) reproduction, (d) sensitivity?
4. Name the taxon that is immediately above the genus.
5. The mule is the offspring of a horse and a donkey. Why does this show that the horse and the donkey are separate species?
6. List three features that distinguish fungi from green plants.
7. A specimen has a moist skin without scales, breathes through gills when young and lungs when adult, and lays eggs in water. Name its class and one example.
8. State two reasons for regarding viruses as non-living and one reason for regarding them as living.
9. An organism has hair, mammary glands and breathes with lungs, but lives in the sea. Name its class.
10. Distinguish between a monocotyledon and a dicotyledon using four features.
11. In a simple experiment on the effect of light on the growth of bean seedlings, what would be (a) the independent variable and (b) a suitable control?

12. Write the scientific name for a plant whose genus is Zea and whose species is mays, observing all the rules for writing it.

Answers to Exercise 1.1

1. Cell, tissue, organ, organ system: cells form tissues, tissues form organs, and organs form systems.
2. Excretion removes the waste products of metabolism, such as urea and carbon dioxide; egestion removes undigested food that never entered the cells. Excreted wastes are made inside cells, but egested material is not.
3. (b) Locomotion from place to place: plants are alive but are usually fixed in position.
4. Family.
5. The mule is sterile, so the horse and donkey cannot interbreed to produce fertile offspring, which is the test of a species.
6. Fungi lack chlorophyll, have walls of chitin instead of cellulose, feed by absorption (saprophytic or parasitic) and store glycogen instead of starch.
7. Amphibia, for example the toad or frog.
8. Non-living: they can be crystallised, they have no cell structure, and they show no life processes outside a host. Living: they reproduce (inside a host cell) and contain genetic material.
9. Mammalia (for example a whale or dolphin): hair, mammary glands and lung breathing outweigh the aquatic habit.
10. Monocots: one cotyledon, parallel venation, fibrous roots, scattered vascular bundles. Dicots: two cotyledons, net venation, tap root, vascular bundles in a ring.
11. (a) The presence or absence of light (or the amount of light). (b) A set of seedlings treated identically in every way, with the same soil, water and

temperature, but kept in the light while the test group is kept in the dark, or the reverse.

12. *Zea mays*: genus with a capital, species in lower case, italicised or underlined.

Chapter summary

- Biology is divided into branches such as botany, zoology, ecology, genetics and microbiology, and uses the scientific method of observation, hypothesis, controlled experiment and conclusion.
- Living things show movement, respiration, sensitivity, growth, reproduction, excretion and nutrition; growth is a permanent increase in dry mass, and excretion differs from egestion.
- Life is organised from cell to tissue, organ, organ system and organism, and then to population, community, ecosystem and biosphere.
- Classification arranges organisms in the sequence kingdom, phylum, class, order, family, genus, species; the species is the basic unit and is made up of organisms that interbreed to give fertile offspring.
- Binomial nomenclature, introduced by Linnaeus, gives every organism a Latin name made of a capitalised genus and a lower-case specific epithet, written in italics.
- The five kingdoms are Monera, Protista, Fungi, Plantae and Animalia; Monera are prokaryotic and the others are eukaryotic.
- Plants are grouped into Bryophyta, Pteridophyta and Spermatophyta; flowering plants are monocotyledons or dicotyledons.

- Animals are grouped into phyla such as Porifera, Cnidaria, Platyhelminthes, Nematoda, Annelida, Mollusca, Arthropoda and Chordata; arthropods include insects, arachnids, crustaceans, centipedes and millipedes, and vertebrates are fishes, amphibians, reptiles, birds and mammals.
- Viruses are acellular obligate parasites with a nucleic acid core and a protein coat, and are on the borderline of life.
- Dichotomous keys identify organisms through paired contrasting statements.

Chapter 2: The Cell and Its Environment

Every living organism, from the smallest bacterium in a pond at Ibadan to the largest baobab in the savanna, is built from cells or is itself a single cell. The cell is the basic structural and functional unit of life. Understanding what a cell contains, how its parts cooperate, how materials enter and leave it, and how it divides to make new cells is the key to understanding almost every later topic in biology, from digestion to heredity.

This chapter describes the cell theory, the light and electron microscopes through which cells are studied, the structure and function of cell components, the difference between plant and animal cells, the specialisation of cells, the movement of substances across membranes, and the process of mitosis.

2.1 The cell theory and the microscope

Cells were first observed in 1665 by the English scientist **Robert Hooke**, who examined thin slices of cork under a simple microscope and saw tiny box-like compartments, which he named cells. In the nineteenth century Matthias Schleiden, a botanist, and Theodor Schwann, a zoologist, concluded that plants and animals are made of cells, and Rudolf Virchow added that new cells arise only from pre-existing cells. These ideas make up the **cell theory**: all living organisms are composed of one or more cells; the cell is the basic unit of structure and function; and all cells arise from existing cells by division.

Most cells are too small to see with the naked eye. The unit used for measuring them is the **micrometre** (μm), one thousandth of a millimetre. A typical animal cell measures about 10 to 30 μm across, and a bacterium is about 1 to 5 μm long. The **light microscope** uses visible light and glass lenses and can magnify up to about 1500 times; it can show the nucleus, the cell wall and chloroplasts, but not the fine detail inside them. The **electron microscope** uses a beam of electrons and magnets, magnifies up to hundreds of thousands of times and has a much higher **resolving power**, which is the ability to distinguish two points that lie close together as separate. It reveals ribosomes, the inner structure of mitochondria and the membranes of the cell, but it can examine only dead and specially prepared material in a vacuum.

The **magnification** of a microscope is calculated by multiplying the power of the eyepiece lens by the power of the objective lens in use. A specimen viewed with a ×10 eyepiece and a ×40 objective is magnified 10 × 40 = 400 times. For a drawing or photograph, the magnification is the size of the image divided by the actual size of the object, with both in the same units.

> **Key point.** Magnification makes an image larger; resolution makes it clearer. Increasing magnification without improving resolution only gives a larger blurred image. The electron microscope is valuable chiefly because of its high resolution.

2.2 Structure of the cell

Cells are of two basic kinds. **Prokaryotic** cells, found in bacteria and cyanobacteria, have no nucleus bounded by a membrane and no membrane-bound organelles. **Eukaryotic** cells, found in all other organisms, have a

nucleus and a range of organelles. Our discussion concerns mainly the eukaryotic cell.

The cell membrane

Every cell is enclosed by a thin **cell membrane**, also called the plasma membrane. It is made of a double layer of phospholipid molecules with protein molecules embedded in it. The membrane is **partially (selectively) permeable**: it allows some substances to pass, such as water, oxygen and carbon dioxide, while restricting others, such as large molecules. It therefore controls what enters and leaves the cell and maintains the internal environment.

Cytoplasm

The **cytoplasm** is the jelly-like material between the membrane and the nucleus. It consists mostly of water, in which enzymes, salts, sugars and amino acids are dissolved. It is the site of many chemical reactions, such as the first stage of respiration, and it holds the organelles in place.

The nucleus

The **nucleus** is usually the most conspicuous organelle. It is surrounded by a double **nuclear envelope** perforated by pores. Inside is **nucleoplasm**, one or more **nucleoli** that make ribosomal material, and **chromatin**, a network of thread-like material composed of DNA and protein. When a cell divides, the chromatin condenses into visible **chromosomes**. The nucleus controls all the activities of the cell and carries the hereditary information. A cell from which the nucleus is removed soon dies, and mature mammalian red blood cells, which lack nuclei, live for only about 120 days.

Mitochondria

Mitochondria (singular mitochondrion) are rod-shaped or oval organelles surrounded by two membranes, the inner of which is folded into **cristae**. They are the sites of aerobic respiration and release energy in the form of ATP, which is why they are called the "powerhouses" of the cell. Cells with high energy demands, such as muscle cells and sperm cells, contain very many mitochondria.

Endoplasmic reticulum, ribosomes and the Golgi apparatus

The **endoplasmic reticulum** (ER) is a network of membrane-bound channels running through the cytoplasm. Rough ER has **ribosomes** attached to its surface and is involved in the synthesis and transport of proteins; smooth ER lacks ribosomes and is involved in the synthesis of lipids. Ribosomes are tiny granules, composed of RNA and protein, where amino acids are joined together to make proteins. They are also found free in the cytoplasm, and they occur in prokaryotes as well. The **Golgi apparatus** is a stack of flattened membranous sacs which modifies, packages and secretes materials such as enzymes and mucus; in plants it contributes to forming the cell wall.

Lysosomes and vacuoles

Lysosomes are small sacs bounded by a single membrane that contain digestive enzymes. They break down worn-out organelles, destroy bacteria engulfed by white blood cells and digest food particles. A **vacuole** is a fluid-filled space surrounded by a membrane. Plant cells typically have one large, permanent central vacuole filled with cell sap, a solution of sugars, salts and pigments, bounded by a membrane called the **tonoplast**. It helps keep the cell

firm (turgid). Animal cells have small, temporary vacuoles; in unicellular freshwater organisms such as *Amoeba* there is a **contractile vacuole** that pumps out excess water.

Structures found only in plant cells

The **cell wall** lies outside the membrane in plant cells. It is made of cellulose, is fully permeable, gives the cell a definite shape and strength, and prevents the cell from bursting when it takes in water. **Chloroplasts** are green, disc-shaped organelles surrounded by two membranes. Inside them are stacks of membranous discs called **grana**, containing the green pigment **chlorophyll**, set in a fluid called the **stroma**. Chloroplasts are the site of photosynthesis. Some plant cells also contain **leucoplasts**, which store starch, and **chromoplasts**, which hold orange and red pigments in fruits and petals. Fungal cells also have cell walls, but these are made of chitin.

Comparing plant and animal cells

Feature	Plant cell	Animal cell
Cell wall	Present (cellulose)	Absent
Shape	Fixed, often regular	Irregular, often rounded
Chloroplasts	Present in green parts	Absent
Vacuole	One large permanent vacuole	Small and temporary, or absent
Stored food	Starch	Glycogen
Centrioles	Absent in higher plants	Present
Nucleus position	Often pushed to the edge by the vacuole	Usually central

Prokaryotes compared with eukaryotes

A bacterial cell has a cell wall (not made of cellulose), a membrane, cytoplasm, ribosomes (smaller than those of eukaryotes) and a single circular chromosome lying in a region called the nucleoid. It has no nuclear envelope, no mitochondria, no ER and no chloroplasts. Many bacteria carry small extra rings of DNA called plasmids and some bear flagella for movement. Prokaryotic cells are generally much smaller than eukaryotic cells.

2.3 Specialisation of cells

In multicellular organisms, cells become specialised in structure so that they can do particular jobs efficiently. This is called **cell differentiation**. Some examples are given in the table below.

Cell	Special feature	Function
Red blood cell	Biconcave disc, no nucleus, contains haemoglobin	Carries oxygen; large surface area for absorption
White blood cell	Flexible shape, lobed or large nucleus	Engulfs microbes or makes antibodies
Muscle cell	Elongated, contractile fibres, many mitochondria	Contraction and movement
Nerve cell (neurone)	Long fibre with branching ends	Conducts impulses
Sperm cell	Tail, streamlined head, many mitochondria, acrosome	Swims to and penetrates the egg
Root hair cell	Long, thin extension of the surface	Absorbs water and mineral salts
Palisade cell	Tall, packed with chloroplasts	Photosynthesis
Guard cell	Kidney-shaped, thick inner wall, chloroplasts	Controls opening of stomata
Xylem vessel	Dead, hollow, lignified walls	Conducts water; supports the plant

Differentiated cells of one type form a tissue. In animals the principal tissues are epithelial, connective, muscular and nervous; in plants they are meristematic (dividing), epidermal, parenchyma, collenchyma, sclerenchyma, xylem and phloem.

2.4 Movement of substances into and out of cells

A cell must take in oxygen, water and nutrients and get rid of wastes. Substances cross the cell membrane by three main processes: diffusion, osmosis and active transport.

Diffusion

Diffusion is the net movement of molecules or ions from a region of higher concentration to a region of lower concentration, down a concentration gradient, as a result of their random motion. It needs no energy from the cell and is therefore a **passive** process. The smell of frying onions spreading through a kitchen, or sugar dissolving evenly in a cup of tea without stirring, are examples. In living things, oxygen diffuses from the air in the alveoli into the blood, and carbon dioxide diffuses the other way.

The rate of diffusion increases when the concentration gradient is steeper, when the temperature is higher, when the surface area is greater, when the diffusion distance is shorter, and when the molecules are smaller. For this reason, exchange surfaces such as the lining of the small intestine and the alveoli are thin and have a large surface area.

Osmosis

Osmosis is the movement of water molecules from a region of higher water potential (a dilute solution) to a region of lower water potential (a concentrated solution) through a partially permeable membrane. Pure water has the highest water potential. Osmosis is a special case of diffusion applied to water.

The effects of osmosis on cells depend on the solution around them. Three terms describe the surrounding solution relative to the cell contents. A **hypotonic** solution is more dilute than the cell contents, an **isotonic** solution has the same concentration and a **hypertonic** solution is more concentrated.

- In a hypotonic solution, water enters the cell. A plant cell swells until the wall resists further expansion, becoming **turgid**; the pressure exerted on

the wall is turgor pressure. An animal cell with no wall, such as a red blood cell, swells and may burst (**lysis**).

- In an isotonic solution there is no net movement of water, and the cell stays the same size. Medical saline of 0.9% sodium chloride is isotonic to human blood.
- In a hypertonic solution, water leaves the cell. A plant cell shrinks and its membrane pulls away from the wall, a condition called **plasmolysis**, so the cell is **flaccid**; an animal cell shrivels (crenation).

Turgor is essential for plants. It keeps herbaceous stems upright and leaves expanded, and it drives the opening of stomata. When the soil around a tomato plant dries out and the cells lose water, the plant wilts. Salting meat or fish, a traditional method of preservation in Nigerian markets, works by osmosis: the concentrated salt draws water out of any bacteria, which are then unable to multiply.

> **Key point.** Water moves by osmosis from the solution with the higher water potential (the less concentrated solution) to the one with the lower water potential (the more concentrated). Students often reverse this; picture water being "pulled" by the stronger solution.

Active transport

Active transport is the movement of substances across a membrane against a concentration gradient, from low to high concentration, using energy from ATP and special carrier proteins in the membrane. It allows cells to accumulate materials they need. Root hair cells absorb mineral ions such as nitrate from the dilute soil solution by active transport; cells of the small intestine absorb glucose when its concentration in the gut falls below that in

the blood; and kidney tubules reabsorb useful substances. Because it needs energy from respiration, active transport slows or stops if oxygen is lacking or if respiratory poisons such as cyanide are present.

Feature	Diffusion	Osmosis	Active transport
Substance moved	Any molecules or ions	Water only	Ions and molecules
Direction	High to low concentration	High to low water potential	Low to high concentration
Energy needed	No	No	Yes, ATP
Membrane required	Not necessarily	Partially permeable	Yes, with carriers

Large particles cannot cross the membrane directly. In **endocytosis** the membrane folds around the particle and takes it inside in a vesicle: *phagocytosis* is the taking in of solid particles, as when white blood cells engulf bacteria, and *pinocytosis* is the taking in of droplets of liquid. **Exocytosis** is the reverse process by which the cell expels materials.

2.5 Cell division: mitosis

Cells divide so that organisms can grow, replace worn-out cells and repair wounds, and in some organisms so that they can reproduce asexually. A cell divides only after it has grown and copied its DNA. The sequence of events from one division to the next is the **cell cycle**, which includes a long period called **interphase**, when the cell grows and the chromosomes are replicated, followed by division of the nucleus (mitosis) and of the cytoplasm (cytokinesis).

Mitosis is the division of a nucleus that produces two daughter nuclei, each genetically identical to the parent nucleus and having the same number of chromosomes. The number of chromosomes in each body cell of a species is constant: 46 in man, 14 in the garden pea, 20 in maize. When the chromosomes become visible at the start of mitosis, each consists of two identical strands called **chromatids**, joined at a region called the **centromere**.

1. **Prophase.** The chromosomes shorten and thicken so that they become visible as paired chromatids. The nucleolus disappears and the nuclear envelope breaks down. In animal cells, the centrioles move to opposite poles and form spindle fibres.
2. **Metaphase.** The chromosomes line up at the equator of the cell, attached by their centromeres to the spindle fibres.
3. **Anaphase.** The centromeres divide, and the spindle fibres pull the separated chromatids, now regarded as chromosomes, to opposite poles of the cell.
4. **Telophase.** The chromosomes reach the poles and lengthen into chromatin again. Nuclear envelopes form around each group, the nucleoli reappear and the spindle disappears.

Cytokinesis follows. In an animal cell the membrane pinches inwards at the equator, forming a furrow that divides the cell in two. In a plant cell, a **cell plate** forms across the middle and develops into a new cell wall.

Mitosis produces genetically identical cells, so it maintains the chromosome number and ensures the continuity of the organism's characteristics. Uncontrolled mitosis is the basis of cancer, in which cells divide without regard to the body's needs and form a tumour. In plants, mitosis takes place

chiefly in the meristems at the tips of roots and shoots, which explains why a student looking at an onion root tip under the microscope can easily find cells in all stages of division.

2.6 Worked examples

WORKED EXAMPLE 2.1

A student draws a cell. The drawing is 60 mm across; the real cell measures 0.03 mm. Calculate the magnification of the drawing. If the student's microscope has a ×10 eyepiece and a ×40 objective, what is the total magnification of the microscope?

Solution: Magnification of the drawing = size of drawing ÷ actual size = 60 ÷ 0.03 = 2000. The drawing is therefore ×2000. The microscope's total magnification is 10 × 40 = ×400. The drawing is larger than the microscope's field magnification because the student has enlarged the drawing further, which is allowed, but it adds no further detail.

WORKED EXAMPLE 2.2

A cell is 25 µm long on a photograph. The photograph measures it as 5 cm. What is the magnification of the photograph?

Solution: Convert to the same units. 5 cm = 50 mm = 50 000 µm. Magnification = 50 000 ÷ 25 = 2000. The photograph is magnified 2000 times.

WORKED EXAMPLE 2.3

Cylinders of yam of equal length are placed in solutions of different concentration. The cylinder in Solution A gains in length, the one in Solution B stays the same length, and the one in Solution C becomes shorter. Interpret the results.

Solution: A cylinder gains length when its cells take in water by osmosis, which happens when the surrounding solution has a higher water potential, that is, is more dilute than the cell sap. Solution A is therefore hypotonic to the yam cells. In B there is no net movement of water, so B is isotonic. A shorter cylinder means water has left the cells, so Solution C is more concentrated than the cell sap, that is, hypertonic. The cells in C are flaccid, and some may be plasmolysed. A concentration between those of A and C, where the change in length is zero, gives the concentration of the cell sap.

WORKED EXAMPLE 2.4

A root hair cell in a Kano farm soil absorbs nitrate ions even though the nitrate concentration in the cell is higher than in the soil water. Which process is responsible? What would happen to the rate of absorption if the soil became waterlogged and short of oxygen?

Solution: Absorption of a substance from a lower to a higher concentration cannot be by diffusion, because diffusion moves substances down a gradient. The process is active transport. It needs ATP, which the cell makes by aerobic respiration. In waterlogged soil there is little oxygen, so respiration, and the supply of ATP, fall. As a result, the rate of absorption of nitrate falls. This is one reason why crops wilt and become yellow when planted in poorly drained fields.

WORKED EXAMPLE 2.5

A body cell of a maize plant contains 20 chromosomes. How many chromosomes will be found in each daughter cell after mitosis? How many chromatids are there in the cell at metaphase?

Solution: Mitosis gives two genetically identical cells with the same chromosome number as the parent, so each daughter cell has 20 chromosomes. At metaphase each of the 20 chromosomes consists of two chromatids, so the cell has $20 \times 2 = 40$ chromatids. After the centromeres divide in anaphase, each chromatid is counted as a separate chromosome, and so the cell briefly has 40 chromosomes before it divides into two cells of 20 each.

2.7 Exercise

Exercise 2.1

1. Who first used the word "cell" and what material did he examine?
2. Name the organelle responsible for (a) aerobic respiration, (b) photosynthesis, (c) protein synthesis, (d) packaging and secretion of materials.
3. List four differences between a plant cell and an animal cell.
4. A cell seen through a ×15 eyepiece and ×40 objective lens is magnified how many times?
5. A cell of actual width 40 µm is drawn 80 mm wide. Calculate the magnification.
6. Define osmosis and explain why a red blood cell placed in distilled water bursts but a plant cell does not.
7. Explain why a lettuce left in cold water becomes crisp, while one sprinkled with a strong salt solution becomes limp.
8. State three differences between diffusion and active transport.
9. State three factors that increase the rate of diffusion across a surface.
10. Arrange in order the stages of mitosis: anaphase, prophase, telophase, metaphase.
11. A human skin cell has 46 chromosomes. State how many chromosomes each daughter cell contains after mitosis, and the number of chromatids in the cell at prophase.
12. Explain why sperm cells and muscle cells contain a large number of mitochondria.

Answers to Exercise 2.1

1. Robert Hooke, in 1665; he examined thin slices of cork.
2. (a) Mitochondrion; (b) chloroplast; (c) ribosome; (d) Golgi apparatus.
3. Plant cells have a cellulose cell wall, chloroplasts, a large permanent vacuole and starch as stored food; animal cells lack these and store glycogen.
4. 15 × 40 = 600 times.
5. 80 mm = 80 000 µm; 80 000 ÷ 40 = 2000, so the magnification is ×2000.
6. Osmosis is the movement of water from a region of higher water potential to one of lower water potential through a partially permeable membrane. In distilled water, both cells take in water; the red blood cell has no wall and bursts, but the plant cell is held by its strong cellulose wall and becomes turgid.
7. In cold water, the cells take in water by osmosis and become turgid, supporting the leaf. In strong salt solution, water leaves the cells by osmosis, so they become flaccid and the leaf wilts.
8. Diffusion needs no energy, but active transport needs ATP; diffusion goes down a concentration gradient, active transport goes against it; active transport requires carrier proteins and stops when respiration stops.
9. Any three of: steeper concentration gradient, higher temperature, larger surface area, shorter distance, smaller molecules.
10. Prophase, metaphase, anaphase, telophase.
11. Each daughter cell has 46 chromosomes. At prophase each chromosome has two chromatids, so 46 × 2 = 92 chromatids.
12. These cells need much energy, for swimming and contraction respectively, and mitochondria release energy by aerobic respiration in the form of ATP.

Chapter summary

- The cell theory states that all organisms are made of cells, that the cell is the unit of structure and function, and that cells arise from existing cells.
- The light microscope magnifies up to about 1500 times; the electron microscope has much higher resolving power. Total magnification equals eyepiece power times objective power.
- Prokaryotic cells lack a nucleus and membrane-bound organelles; eukaryotic cells have both.
- The nucleus controls the cell and carries chromosomes; mitochondria release energy; ribosomes make protein; the Golgi apparatus packages secretions; lysosomes digest; chloroplasts photosynthesise; the cellulose wall supports plant cells.
- Cells are specialised in structure to perform particular functions, for example red blood cells, root hair cells and palisade cells.
- Diffusion and osmosis are passive; active transport uses ATP to move substances against a concentration gradient.
- In hypotonic surroundings, plant cells become turgid and animal cells may burst; in hypertonic surroundings, plant cells become plasmolysed and animal cells shrink.
- Mitosis, consisting of prophase, metaphase, anaphase and telophase, gives two genetically identical daughter nuclei and serves growth, repair and asexual reproduction.

Chapter 3: Nutrition and Digestion

Every living organism needs a continuous supply of materials and energy. These are obtained through **nutrition**, the process by which an organism acquires and uses food for growth, repair, energy release and the regulation of its activities. Green plants manufacture their own food from simple inorganic substances; animals, fungi and most bacteria must obtain ready-made organic matter. This chapter examines both patterns of feeding, the classes of food substances and the diseases caused by their lack, and the way the human alimentary canal converts complex food into small soluble molecules that the body can absorb.

Nutrition is a favourite area of the UTME. Candidates are regularly asked to identify the product of a particular enzyme, to match a deficiency disease with a missing nutrient, to name the site of absorption, or to interpret the results of a food test. The treatment below is organised so that each of these question types can be answered from first principles rather than by memory alone.

3.1 Modes of nutrition

Organisms are grouped according to the source of their carbon and energy. An **autotroph** builds complex organic molecules from simple inorganic ones such as carbon dioxide and water. A **heterotroph** depends on organic compounds already made by other organisms.

Autotrophic nutrition

Autotrophs fall into two groups. **Photoautotrophs**, such as green plants, algae and cyanobacteria, use light energy trapped by chlorophyll to carry out photosynthesis. **Chemoautotrophs**, such as the nitrifying bacteria *Nitrosomonas* and *Nitrobacter*, obtain energy from the oxidation of inorganic substances like ammonium compounds and nitrites, and use that energy to build organic molecules. Chemosynthesis is much rarer than photosynthesis but is essential in the nitrogen cycle, as Chapter 8 explains.

Heterotrophic nutrition

Heterotrophs include several distinct feeding patterns:

- **Holozoic nutrition**: the ingestion of solid food followed by internal digestion, absorption and assimilation. Humans, goats, lizards and the housefly are holozoic. Holozoic animals are further described as herbivores (plant eaters, such as cattle in Sokoto), carnivores (flesh eaters, such as the lion), omnivores (both, such as humans and the pig) and detritivores (feeders on dead organic debris, such as earthworms and millipedes).
- **Saprophytic nutrition**: the secretion of digestive enzymes onto dead organic matter outside the body, followed by absorption of the soluble products. Bread mould (*Rhizopus*), mushrooms and many bacteria of decay are saprophytes. They are the principal decomposers in ecosystems.
- **Parasitic nutrition**: a close association in which the parasite obtains food from a living host and usually harms it. Examples are the tapeworm in the intestine of man, the mistletoe on a mango tree and the malaria parasite, *Plasmodium*, in the red blood cells.

- **Symbiosis** (mutualism): an association in which both partners benefit. The nitrogen-fixing bacterium *Rhizobium* lives in nodules on the roots of beans and groundnut; it receives carbohydrate from the plant and supplies nitrates in return. Lichens are an association of an alga, which photosynthesises, and a fungus, which gives protection and absorbs water and minerals.
- **Commensalism**: one partner benefits and the other is neither helped nor harmed, as with an orchid growing on the trunk of an oil palm.

> **Key point.** A parasite harms its host, a mutualist helps it, and a commensal does neither. A saprophyte feeds on dead matter, whereas a parasite feeds on living tissue. Many candidates confuse these two; remember that "sapro" means rotten.

3.2 Photosynthesis in brief

Photosynthesis is the foundation of almost every food chain, and it is treated fully in the chapter on plants. Here we need only the summary equation and the main requirements. Carbon dioxide and water, in the presence of light and chlorophyll, yield glucose and oxygen:

$$6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2$$

The glucose is converted to starch for storage, to sucrose for transport, and to cellulose, lipids and, with nitrates and other mineral ions, to amino acids and proteins. Light energy is thus converted to the chemical energy of food, which then passes through the feeding relationships described in the ecology chapter.

3.3 Classes of food and their functions

A **balanced diet** contains all the food classes in the correct proportions, together with enough water and roughage to keep the body healthy. There are seven: carbohydrates, proteins, lipids (fats and oils), vitamins, mineral salts, water and dietary fibre.

Carbohydrates

Carbohydrates contain carbon, hydrogen and oxygen, with hydrogen and oxygen in the ratio 2 : 1, as in water. The simplest are the **monosaccharides** such as glucose, fructose and galactose ($C_6H_{12}O_6$). Two monosaccharides join, with loss of a water molecule, to form a **disaccharide**: maltose (glucose + glucose), sucrose (glucose + fructose) and lactose (glucose + galactose). Many monosaccharides form a **polysaccharide**: starch and cellulose in plants, glycogen in animals. The main function of carbohydrates is the supply of energy; a gram of carbohydrate releases about 17 kJ. Glycogen is stored in the liver and muscles; cellulose forms the cell wall and gives roughage.

Proteins

Proteins contain carbon, hydrogen, oxygen and nitrogen, and often sulphur. They are long chains of **amino acids** joined by peptide bonds. Twenty amino acids occur in proteins, of which eight cannot be made by the adult body and must be supplied in the diet; these are the **essential amino acids**. Proteins are needed for growth and repair, for making enzymes, antibodies, haemoglobin and some hormones, and, when carbohydrates and fats are lacking, for energy (about 17 kJ per gram). First-class (complete) proteins, found in meat, fish, eggs, milk and soya beans, contain all the essential amino acids. Second-class

proteins, such as those in maize and beans taken separately, lack one or more, which is why a mixed diet of beans together with a cereal such as maize or rice is nutritionally sound.

Lipids

Fats and oils consist of carbon, hydrogen and oxygen, with much less oxygen than carbohydrates. A lipid molecule is built from one molecule of glycerol and three of fatty acid. Fats are solid at room temperature (butter, lard); oils are liquid (palm oil, groundnut oil). Lipids provide the most concentrated source of energy, about 39 kJ per gram, form an insulating layer beneath the skin, protect organs such as the kidney, are components of cell membranes, and act as solvents for vitamins A, D, E and K.

Vitamins

Vitamins are organic compounds needed in very small amounts for normal metabolism. They are not sources of energy. Fat-soluble vitamins are A, D, E and K; water-soluble ones are the B group and C.

Vitamin	Source	Function	Deficiency disease
A (retinol)	Liver, palm oil, carrot, milk	Healthy epithelia; formation of visual pigment	Night blindness, xerophthalmia
B_1 (thiamine)	Unpolished rice, yeast, beans	Carbohydrate respiration; nerve function	Beriberi
B_3 (niacin)	Meat, groundnut, whole grain	Respiration	Pellagra
C (ascorbic acid)	Orange, guava, pawpaw, tomato	Healthy gums, blood vessels, wound healing	Scurvy
D (calciferol)	Fish oil, egg yolk; made in skin in sunlight	Absorption of calcium and phosphorus	Rickets (children), osteomalacia (adults)
K	Green vegetables; made by gut bacteria	Blood clotting (prothrombin)	Excessive bleeding

Mineral salts

Minerals are inorganic ions needed in small but essential amounts. Calcium and phosphorus build bones and teeth, and calcium also helps blood to clot and muscles to contract. Iron is a component of haemoglobin; its lack causes **anaemia**. Iodine is needed to make thyroxine; its deficiency causes **goitre**, a swelling of the thyroid gland in the neck. Sodium and potassium maintain nerve action and water balance. Fluoride hardens tooth enamel. Magnesium is a component of chlorophyll in plants.

Water and roughage

Water makes up about 65 to 70 per cent of the human body. It is the solvent in which chemical reactions occur, the medium of transport in blood, a reactant in hydrolysis and photosynthesis, and the means of temperature regulation through sweating. Roughage, or dietary fibre, consists chiefly of cellulose. It is not digested but gives bulk to the food, stimulates peristalsis and prevents constipation.

Malnutrition

Malnutrition is any condition caused by a diet that is lacking, in excess or unbalanced in some respect. It includes undernourishment (too little food), over-nutrition (obesity) and deficiency diseases. Two protein-related disorders are common in children in poor communities. **Kwashiorkor** results from a severe lack of protein even when energy intake may be adequate; the child has a swollen abdomen and legs (oedema), thin limbs, pale hair and skin changes. **Marasmus** results from a shortage of both protein and energy; the child is severely wasted and looks like "skin and bone". Treatment is by gradual introduction of a balanced diet rich in protein, such as milk, beans, fish and eggs.

> **Key point.** Kwashiorkor: protein deficiency, swollen belly. Marasmus: overall food and energy deficiency, wasting. Obesity, diabetes and heart disease are diseases of excess or imbalance, and are also counted as malnutrition.

3.4 Food tests

Laboratory tests identify the class of food present in a sample. The principal tests are summarised below.

Food	Test	Positive result
Starch	Add a few drops of iodine solution	Blue-black colour
Reducing sugar (glucose, maltose, lactose)	Heat with Benedict's (or Fehling's) solution	Blue turns green, yellow, orange, then brick-red precipitate
Non-reducing sugar (sucrose)	Boil with dilute hydrochloric acid, neutralise with sodium hydrogencarbonate, then heat with Benedict's solution	Brick-red precipitate (negative before hydrolysis)
Protein	Biuret test: sodium hydroxide, then a few drops of dilute copper(II) sulphate	Violet or purple colour
Protein (alternative)	Millon's test	Red or pink precipitate
Fat or oil	Ethanol emulsion test: shake with ethanol, then pour into water; or rub on paper	Cloudy white emulsion; or a translucent grease spot
Vitamin C	Add to DCPIP solution	Blue DCPIP is decolourised

Sucrose is a non-reducing sugar, so it gives no colour change with Benedict's solution until it has been hydrolysed by acid into glucose and fructose, which are reducing sugars. A sample that gives a negative Benedict's test but a positive result after acid hydrolysis therefore contains sucrose.

WORKED EXAMPLE 3.1

A food extract is divided into three portions. Iodine solution gives no colour change. Heating with Benedict's solution gives no change. The portion boiled with dilute acid, neutralised and heated with Benedict's solution gives a brick-red precipitate. Which food class is present?

Solution: The absence of a blue-black colour rules out starch. The negative Benedict's test shows that no reducing sugar was present. After acid hydrolysis a reducing sugar appears, so the original substance must have been a non-reducing sugar that was broken down into reducing sugars. The extract contains sucrose.

3.5 Calculating the energy value of food

The energy in food is expressed in kilojoules (kJ). It can be estimated from the mass of each class using the approximate values of 17 kJ per gram for carbohydrate and protein and 39 kJ per gram for fat. A simple experiment burns a dry weighed food sample under a boiling tube of water and measures the temperature rise. The energy gained by the water is mass × specific heat capacity × temperature rise, with 4.2 J per gram per °C for water.

WORKED EXAMPLE 3.2

A meal contains 80 g of carbohydrate, 30 g of protein and 20 g of fat. Calculate its total energy value, using 17 kJ/g for carbohydrate and protein and 39 kJ/g for fat.

Solution: Carbohydrate: 80 × 17 = 1,360 kJ. Protein: 30 × 17 = 510 kJ. Fat: 20 × 39 = 780 kJ. Total = 1,360 + 510 + 780 = 2,650 kJ.

WORKED EXAMPLE 3.3

In an experiment, 1.0 g of groundnut burns and raises the temperature of 20 g of water from 25 °C to 85 °C. Estimate the energy released per gram, taking 4.2 J/g/°C as the specific heat capacity of water and ignoring losses.

Solution: Temperature rise = 85 − 25 = 60 °C. Energy = 20 × 4.2 × 60 = 5,040 J = 5.04 kJ per gram. The real value is much higher (groundnut is rich in fat), which shows that a large part of the heat was lost to the surroundings in such a simple apparatus.

3.6 Dentition

Teeth break food into small pieces, increasing the surface area for enzyme action. A tooth has a crown above the gum, a neck at the gum and a root fixed in a socket of the jaw bone. The crown is covered with **enamel**, the hardest substance in the body; beneath it is **dentine**, a bone-like material; the central **pulp cavity** holds blood vessels and nerves. The root is covered with cement, which fixes it to the jaw by fibres.

Mammals have four types of teeth, a condition called **heterodont**: incisors for cutting and biting, canines for tearing, premolars and molars for crushing and grinding. Humans have two sets: 20 milk (deciduous) teeth in childhood, replaced by 32 permanent teeth in adulthood. The adult dental formula is written for one half of the upper and lower jaw:

i 2/2, c 1/1, pm 2/2, m 3/3 = 8 per half, so 8 × 4 = 32 in all, where the figure above the line is for the upper jaw and the figure below for the lower.

The diet of an animal is reflected in its teeth. Herbivores such as the cow and the goat have broad grinding molars and incisors only in the lower jaw, which bite against a hard pad in the upper jaw; canines are absent or reduced, leaving a gap called a diastema. Carnivores such as the dog and the lion have large pointed canines and specialised **carnassial** teeth (the last upper premolar and first lower molar) that shear flesh like scissors.

Tooth decay (dental caries) begins when bacteria in the mouth feed on sugar and produce acid, which dissolves the enamel and then the dentine. It is prevented by brushing after meals, reducing sugar, adequate calcium and fluoride, and regular dental visits.

3.7 The human alimentary canal

Digestion is the breakdown of large insoluble food molecules into small soluble ones that can pass through the walls of the gut into the blood. It involves **mechanical digestion** (chewing, churning of the stomach) and **chemical digestion**, which is the hydrolysis of food by enzymes. The alimentary canal is a muscular tube extending from the mouth to the anus. Its parts, in order, are the mouth, pharynx, oesophagus, stomach, small intestine (duodenum, jejunum and ileum), large intestine (colon and rectum) and anus. The salivary glands, liver and pancreas are associated glands that pour secretions into the canal.

Food moves along the canal by **peristalsis**, waves of rhythmic contraction of the circular and longitudinal muscles in its wall. Peristalsis is the reason why one can swallow food even when standing on one's head.

The mouth

Teeth chew the food, which is mixed with saliva from three pairs of salivary glands. Saliva contains water, mucus and the enzyme **salivary amylase** (ptyalin), which acts best at about pH 7 and hydrolyses starch to maltose. The tongue rolls the chewed food into a ball, the **bolus**, and pushes it to the back of the throat. During swallowing the epiglottis closes the opening of the windpipe, so food passes into the oesophagus and not into the lungs.

The stomach

The stomach is a muscular bag whose churning movements mix the food with gastric juice into a creamy liquid called **chyme**. Gastric juice contains hydrochloric acid, which gives a pH of about 2, kills most bacteria and provides the acid medium needed by pepsin. The enzyme **pepsin**, secreted as inactive pepsinogen, hydrolyses proteins to shorter chains of amino acids called polypeptides. In infants, **rennin** curdles the milk protein casein so that it stays longer in the stomach. Mucus protects the stomach wall from being digested by its own acid and enzyme; failure of this protection leads to ulcers. The stomach is separated from the intestine by a ring of muscle, the pyloric sphincter, which releases the chyme gradually.

The small intestine

The duodenum, the first 25 cm or so of the small intestine, receives bile from the liver and gallbladder and pancreatic juice from the pancreas. The ileum completes digestion and is the main region of absorption.

- **Bile** is made in the liver and stored in the gallbladder. It contains no enzyme. It is alkaline (neutralising the acid chyme and giving the right pH for pancreatic and intestinal enzymes), and its bile salts **emulsify** fats, breaking large globules into tiny droplets with a large surface area on

which lipase can act. Bile also contains bile pigments from the breakdown of haemoglobin.

- **Pancreatic juice** contains **pancreatic amylase** (starch to maltose), **trypsin** (proteins and polypeptides to shorter peptides) and **lipase** (fats to fatty acids and glycerol), together with sodium hydrogencarbonate, which makes the juice alkaline.
- The wall of the ileum secretes **intestinal juice** (succus entericus) containing **maltase** (maltose to glucose), **sucrase** (sucrose to glucose and fructose), **lactase** (lactose to glucose and galactose), **peptidases** (peptides to amino acids; erepsin) and lipase.

Region	Enzyme	Substrate	Products
Mouth	Salivary amylase	Starch	Maltose
Stomach	Pepsin (and rennin in infants)	Protein (casein)	Polypeptides
Duodenum	Pancreatic amylase	Starch	Maltose
Duodenum	Trypsin	Proteins, polypeptides	Peptides
Duodenum	Lipase	Emulsified fats	Fatty acids and glycerol
Ileum	Maltase	Maltose	Glucose
Ileum	Sucrase	Sucrose	Glucose and fructose
Ileum	Lactase	Lactose	Glucose and galactose
Ileum	Peptidases	Peptides	Amino acids

Key point. End products of digestion: carbohydrates give simple sugars (mainly glucose); proteins give amino acids; fats give fatty acids and glycerol. Cellulose is not digested by humans because we lack cellulase.

Absorption

Absorption is the passage of digested food into the blood or lymph. It occurs mainly in the ileum, which is adapted by being long (about 5 to 6 m in an adult), by folding of its inner lining, and by thousands of finger-like **villi** whose epithelial cells carry microvilli. Each villus has a thin one-cell-thick epithelium, a dense capillary network and a central lymph vessel called a **lacteal**. Glucose and amino acids pass into the capillaries, by diffusion and active transport, and are carried in the hepatic portal vein to the liver. Fatty acids and glycerol enter the epithelial cells, recombine into fats, and pass mainly into the lacteals, from where lymph carries them into the blood near the heart. Vitamins and mineral salts are absorbed along with them; much water is absorbed in the small intestine too.

The large intestine

The colon absorbs most of the remaining water and some mineral salts, and the undigested material, consisting of cellulose, dead cells, bacteria and bile pigments, becomes the semi-solid **faeces**. These are stored in the rectum and expelled through the anus in egestion. Bacteria in the colon synthesise vitamin K and some B vitamins. Note that *egestion* (removal of undigested food) is different from *excretion* (removal of metabolic waste).

Assimilation and the roles of the liver

The liver receives all blood from the intestine through the hepatic portal vein and controls what passes on to the body. Excess glucose is converted to glycogen under the influence of insulin; when blood sugar falls, glycogen is changed back to glucose under the influence of glucagon. Amino acids are used to make plasma proteins; excess amino acids are **deaminated**, the amino group being converted to urea for excretion and the remainder used as an energy source or converted to fat. The liver also stores vitamins A, D and B_{12} and iron, makes bile, detoxifies drugs and alcohol, and produces heat. **Assimilation** is the use of absorbed food by cells for energy, growth and repair.

WORKED EXAMPLE 3.4

A plate of boiled yam and egg sauce containing starch, protein and oil passes through the gut. State the enzymes responsible and the end products for each food class.

Solution: Starch in the yam is hydrolysed by salivary and pancreatic amylase to maltose, and maltase in the ileum converts this to glucose. Protein in the egg is acted on by pepsin in the stomach, trypsin in the duodenum and peptidases in the ileum, and the end products are amino acids. The oil is emulsified by bile salts and then hydrolysed by lipase to fatty acids and glycerol.

3.8 Digestion in other animals

Ruminants such as cattle, sheep and goats feed on grass rich in cellulose. Their stomach has four chambers: rumen, reticulum, omasum and abomasum. The rumen contains enormous numbers of bacteria and protozoa which secrete cellulase and digest cellulose to simple sugars and fatty acids. The animal regurgitates the partly digested food as **cud**, chews it again and swallows it, a process called rumination. The abomasum is the true stomach, where gastric juice acts. Herbivores that do not ruminate, such as the rabbit, also depend on bacteria, in a long caecum and appendix.

In a simple unicellular organism such as *Amoeba*, food is engulfed by pseudopodia into a **food vacuole**; digestive enzymes from the cytoplasm enter the vacuole, soluble products diffuse into the cytoplasm, and undigested remains are egested at the cell surface. This is intracellular digestion. In the earthworm, food passes into a muscular gizzard where grit grinds it finely, and digestion is extracellular within the intestine.

3.9 Common disorders of the digestive system

- **Constipation**: hard, infrequent stools, usually from low roughage or low water intake.
- **Diarrhoea**: frequent watery stools, often from infection by bacteria, viruses or protozoa in contaminated water; it causes dangerous loss of water and salts, and is treated with oral rehydration solution (salt, sugar and clean water).
- **Peptic ulcer**: an open sore on the lining of the stomach or duodenum, often associated with the bacterium *Helicobacter pylori*, excess acid, stress or heavy use of some painkillers.

- **Appendicitis**: inflammation of the appendix, which if untreated may burst.
- **Jaundice**: yellowing of the skin and eyes through a build-up of bile pigment, caused by liver disease or blockage of the bile duct.
- **Obesity**: excess body fat from taking in more energy than is used.

Exercise 3.1

1. Which of the following is a chemoautotroph? (a) Maize (b) *Nitrobacter* (c) *Rhizopus* (d) The tapeworm
2. Name the end products of digestion of (a) starch, (b) protein and (c) fat.
3. A lipid sample is shaken with ethanol and poured into water. What is observed if fat is present?
4. Which vitamin deficiency produces rickets, and which mineral is poorly absorbed as a result?
5. A snack contains 50 g of carbohydrate, 10 g of protein and 25 g of fat. Calculate its energy value, using 17 kJ/g for carbohydrate and protein and 39 kJ/g for fat.
6. Give two reasons why bile, although it contains no enzyme, is important for digestion.
7. Write the adult human dental formula and calculate the total number of teeth.
8. Explain why the ileum is well adapted for absorption.
9. In an experiment, 0.5 g of dry biscuit raises the temperature of 40 g of water by 20 °C. Calculate the energy gained by the water in kilojoules (specific heat capacity of water 4.2 J/g/°C).
10. A solution gives a violet colour with sodium hydroxide and dilute copper sulphate. What does it contain?

11. Distinguish between kwashiorkor and marasmus.
12. Why is it advantageous for a ruminant to chew the cud?

Answers to Exercise 3.1

1. (b) *Nitrobacter*: it gets energy from oxidising nitrites, not from light.
2. (a) Glucose (via maltose); (b) amino acids; (c) fatty acids and glycerol.
3. A cloudy white emulsion forms, because the fat dissolves in ethanol and is dispersed as tiny droplets in water.
4. Vitamin D; calcium (and phosphorus), because vitamin D promotes their absorption.
5. 50 × 17 = 850; 10 × 17 = 170; 25 × 39 = 975; total = 850 + 170 + 975 = 1,995 kJ.
6. It neutralises acid chyme to provide the alkaline medium for pancreatic enzymes, and its salts emulsify fats, increasing the surface area for lipase.
7. i 2/2, c 1/1, pm 2/2, m 3/3; (2 + 1 + 2 + 3) × 2 = 16 per jaw, so 32 teeth in all.
8. It is long, with folds, villi and microvilli giving a very large surface; the epithelium is one cell thick; each villus has a rich capillary network and a lacteal that maintain a steep concentration gradient.
9. Energy = 40 × 4.2 × 20 = 3,360 J = 3.36 kJ.
10. Protein (positive biuret test).
11. Kwashiorkor is mainly protein deficiency, with oedema and a swollen belly; marasmus is a deficiency of both food energy and protein, with severe wasting.

12. Chewing again breaks plant cell walls into finer pieces, increasing the surface area for attack by cellulase-secreting microbes, so digestion of cellulose is more complete.

Chapter summary

- Autotrophs make their own food (photoautotrophs use light, chemoautotrophs use chemical energy); heterotrophs may be holozoic, saprophytic, parasitic, symbiotic or commensal.
- The seven classes of food are carbohydrates, proteins, lipids, vitamins, minerals, water and roughage; energy values are about 17 kJ/g for carbohydrate and protein and 39 kJ/g for fat.
- Deficiency diseases include scurvy (C), rickets (D), beriberi (B_1), night blindness (A), anaemia (iron), goitre (iodine), kwashiorkor (protein) and marasmus (food energy).
- Food tests: iodine for starch, Benedict's for reducing sugar, biuret for protein, ethanol emulsion for fats, DCPIP for vitamin C.
- Human adult dentition is 2.1.2.3 in each half jaw, giving 32 teeth; herbivores and carnivores show different adaptations.
- Digestion proceeds by amylases, pepsin, trypsin, lipase, maltase, sucrase, lactase and peptidases to glucose, amino acids, fatty acids and glycerol.
- Absorption occurs mainly in the ileum through villi; the colon absorbs water; the liver regulates blood sugar and deaminates amino acids.
- Ruminants digest cellulose with the help of microbes in the rumen.

Chapter 4: Transport Systems and Gaseous Exchange

In a single-celled organism such as *Amoeba*, every part of the cell is close to the surrounding water, and oxygen, food and wastes move in and out by diffusion alone. In a large multicellular organism, most cells are far from the outside world, and diffusion over such distances would be hopelessly slow. A transport system is therefore needed to carry nutrients, gases, hormones and wastes between the exchange surfaces and the cells. This chapter describes transport in mammals and in flowering plants, and the organs by which gases are exchanged with the environment.

The topics here appear in the UTME every year: the structure of the heart and the path of blood through it, the composition and functions of blood, the ABO blood groups, the movement of water in the xylem, the mechanism of stomatal opening, and the way the lungs and gills are adapted to their task. We take them in turn.

4.1 Why transport systems are needed

The need for a transport system rises with body size, because the ratio of surface area to volume falls as an organism grows. Consider two cubes. A cube of side 1 cm has a surface area of 6 cm² and a volume of 1 cm³, so its ratio is 6 : 1. A cube of side 4 cm has a surface area of 6 × 16 = 96 cm² and a volume of 64 cm³, giving a ratio of 1.5 : 1. The larger body has far less surface per unit of volume, and its interior cells cannot be supplied by diffusion alone. Large animals therefore have a heart and blood vessels, and large plants have xylem and phloem.

4.2 Blood

An adult human has roughly 5 litres of blood, which is about 7 to 8 per cent of the body mass. Blood is a tissue made of a liquid, the **plasma**, and cells and cell fragments suspended in it.

Plasma

Plasma makes up about 55 per cent of blood by volume and is 90 per cent water. It carries dissolved glucose, amino acids, mineral ions, vitamins, hormones, urea and carbon dioxide (mostly as hydrogencarbonate ions), and also the plasma proteins: albumin, globulins (including antibodies) and fibrinogen. Plasma from which fibrinogen has been removed by clotting is called **serum**.

Red blood cells

Red blood cells (erythrocytes) are biconcave discs with no nucleus when mature. They number about 5 million per cubic millimetre of blood, are made in the red bone marrow, and live for about 120 days before being broken down in the liver and spleen. Their cytoplasm is packed with **haemoglobin**, a protein containing iron that combines reversibly with oxygen:

$Hb + 4O_2 \rightleftharpoons Hb(O_2)_4$ (haemoglobin ⇌ oxyhaemoglobin)

In the lungs, where oxygen is abundant, the reaction goes to the right and the blood becomes bright red; in respiring tissues, where oxygen is scarce, oxyhaemoglobin releases its oxygen. The biconcave shape gives a large surface area for gas exchange, and the lack of a nucleus leaves more room for haemoglobin. Carbon monoxide from car exhaust or poor charcoal stoves combines with haemoglobin about 200 times more strongly than oxygen does, which is why it is so dangerous.

White blood cells

White blood cells (leucocytes) have nuclei, are larger than red cells and are fewer, about 7,000 per cubic millimetre. They defend the body. **Phagocytes** (neutrophils and monocytes) engulf bacteria by phagocytosis. **Lymphocytes** make antibodies, which bind to the antigens on invading microbes and mark them for destruction or neutralise toxins.

Platelets

Platelets (thrombocytes) are small fragments of cells, without nuclei, that help blood to clot. When a vessel is cut, platelets and damaged tissue release thromboplastin, which in the presence of calcium ions and vitamin K helps convert prothrombin to the enzyme thrombin. Thrombin changes soluble fibrinogen to insoluble fibrin threads, which form a mesh trapping blood cells to make the clot. Clotting prevents excessive loss of blood and entry of germs. The inherited disorder **haemophilia**, discussed in the chapter on genetics, is a failure of clotting.

Component	Main function
Plasma	Transport of dissolved food, hormones, urea, CO_2; distribution of heat
Red blood cells	Transport of oxygen as oxyhaemoglobin
White blood cells	Phagocytosis; antibody production
Platelets	Blood clotting

Blood groups

The surface of red cells may carry the antigens A or B. The ABO blood group of a person is determined by these antigens, and the plasma contains antibodies against any antigen the red cells lack.

Group	Antigen on red cells	Antibody in plasma	Can donate to	Can receive from
A	A	anti-B	A, AB	A, O
B	B	anti-A	B, AB	B, O
AB	A and B	none	AB	A, B, AB, O
O	none	anti-A and anti-B	A, B, AB, O	O

Group O is called the **universal donor**, since its red cells carry no antigen that the recipient's antibodies can attack, and group AB the **universal recipient**, since it has no antibodies in the plasma. If incompatible blood is transfused, antibodies cause the donor's red cells to clump together (agglutination), blocking small vessels with dangerous results. Blood is also grouped by the **rhesus factor**: people with the Rh antigen are Rh-positive, and those without are Rh-negative. Most Nigerians are Rh-positive. A problem can arise when an Rh-negative mother carries a second Rh-positive baby, because antibodies she made against the first child's cells can cross the placenta and attack the red cells of the second child.

> **Key point.** The antigens are on the red cells; the antibodies are in the plasma. A person cannot carry an antibody against his or her own antigen. This single rule lets you rebuild the whole table.

4.3 The heart and circulation

The human circulatory system is a **double circulation** in which blood passes through the heart twice in each complete circuit of the body: once through the pulmonary circuit, from the heart to the lungs and back, and once through the systemic circuit, from the heart to the body and back. It is a **closed** system because blood stays inside vessels, unlike the open system of insects, where blood bathes the tissues directly in a body cavity called the haemocoel.

Structure of the heart

The heart is a muscular organ of cardiac muscle lying between the lungs, slightly to the left. It has four chambers. The two upper chambers, the **atria** (right and left), receive blood; the two lower chambers, the **ventricles**, pump it out. A wall called the septum keeps oxygenated blood in the left side apart from deoxygenated blood on the right.

- The **right atrium** receives deoxygenated blood from the body through the **vena cava** (superior and inferior).
- Blood passes through the **tricuspid valve** to the **right ventricle**, which pumps it through the **pulmonary artery** to the lungs. This is the only artery in the body that carries deoxygenated blood.
- The **left atrium** receives oxygenated blood from the lungs through the **pulmonary veins**, the only veins that carry oxygenated blood.
- It passes through the **bicuspid (mitral) valve** to the **left ventricle**, which pumps blood through the **aorta** to the whole body.

The wall of the left ventricle is much thicker than that of the right, because it must generate enough pressure to push blood around the entire body, whereas the right ventricle needs only to drive blood through the nearby lungs. The

valves, together with semilunar valves at the base of the aorta and the pulmonary artery, keep blood moving in one direction. The characteristic "lub-dub" sounds of the heart are produced by the closing of these valves. The heart muscle itself is supplied by the **coronary arteries**; their blockage by fatty deposits causes a heart attack.

The cardiac cycle

A heartbeat has three phases. In **atrial systole** both atria contract and push blood into the ventricles. In **ventricular systole** the ventricles contract, the atrioventricular valves slam shut and blood is forced out into the arteries. In **diastole** the whole heart relaxes and the atria refill. The resting pulse of an adult is around 70 beats per minute, and each beat of the left ventricle pushes out about 70 cm³ of blood. The **cardiac output**, the volume of blood pumped by the heart each minute, is therefore the product of the heart rate and the stroke volume. The pacemaker, a patch of specialised muscle called the sino-atrial node in the wall of the right atrium, starts each beat.

WORKED EXAMPLE 4.1

A man at rest has a heart rate of 72 beats per minute and a stroke volume of 75 cm³. Calculate his cardiac output in litres per minute. During exercise his rate rises to 150 beats per minute and the stroke volume to 100 cm³. By how many times has the output increased?

Solution: At rest: 72 × 75 = 5,400 cm³/min = 5.4 litres/min. During exercise: 150 × 100 = 15,000 cm³/min = 15 litres/min. The increase is 15 ÷ 5.4 = 2.78, so the output is about 2.8 times greater.

Blood vessels

There are three kinds of vessel, each adapted to its job.

Feature	Artery	Vein	Capillary
Direction of flow	Away from the heart	Towards the heart	Between arteries and veins
Wall	Thick, muscular and elastic	Thin, less muscle	One cell thick
Lumen	Narrow	Wide	Very narrow
Valves	None (except at the heart exits)	Present	None
Pressure	High, pulsing	Low, steady	Falling
Typical blood	Oxygenated (except pulmonary)	Deoxygenated (except pulmonary)	Changes by exchange

Arteries have elastic walls that stretch with each surge of blood and recoil to keep it moving; the pulse is felt where an artery crosses a bone, such as at the wrist. Veins depend on the squeezing action of skeletal muscles and on valves to prevent backflow. Capillaries form dense networks in tissues; their thin walls allow oxygen, glucose and other small molecules to leave the blood, and carbon dioxide and wastes to enter it. Blood pressure is measured by a sphygmomanometer and written as systolic over diastolic pressure, a healthy young adult having about 120/80 mm of mercury.

Tissue fluid and lymph

At the arterial end of a capillary the pressure forces plasma, without most of its proteins and without red cells, through the thin wall to form **tissue fluid** bathing the cells. Much of it returns to the capillary at the venous end by osmosis; the rest drains into **lymph vessels** as **lymph**. Lymph is moved by

muscle action through lymph nodes, where lymphocytes multiply and bacteria are filtered, and it empties into veins near the heart. The lymphatic system also carries absorbed fats from the villi.

Circulatory disorders

- **Hypertension**: persistently raised blood pressure, linked to salt, stress, obesity and smoking.
- **Atherosclerosis**: deposition of cholesterol in arterial walls, narrowing them.
- **Thrombosis**: formation of a clot inside a vessel; if in a coronary artery it causes a heart attack, and in the brain a stroke.
- **Anaemia**: a shortage of haemoglobin or red cells, often from lack of iron.
- **Sickle-cell anaemia**: an inherited condition in which abnormal haemoglobin distorts the red cells; discussed in Chapter 6.

4.4 Transport in flowering plants

Flowering plants have two kinds of conducting tissue, grouped together in vascular bundles. **Xylem** carries water and dissolved mineral salts from the roots upward. **Phloem** carries manufactured food, chiefly sucrose, from the leaves to wherever it is used or stored.

Absorption of water and minerals

Water enters the plant through the **root hairs**, which are extensions of epidermal cells that greatly increase the surface area. The cell sap of the root hair has a lower water potential (a more concentrated solution) than the soil water, so water enters by osmosis. It then passes across the cortex, through the cells or between their walls, into the xylem in the centre of the root. Mineral

ions enter by diffusion and, more importantly, by active transport against the concentration gradient, using energy from respiration; waterlogged soil, in which there is no oxygen for respiration, therefore reduces mineral uptake.

Xylem and the movement of water upward

Xylem vessels are long, dead, hollow tubes with walls strengthened by lignin; they have no cross walls and so offer little resistance to flow. Three forces move water upward, even to the top of tall trees:

- **Transpiration pull**: as water evaporates from the leaf, the loss draws more water up the xylem in a continuous stream.
- **Cohesion and adhesion**: water molecules stick to each other (cohesion), so the column does not break, and to the walls of the vessels (adhesion).
- **Root pressure**: osmotic entry of water into the roots pushes it up the stem for a short distance. It is important at night or in humid conditions but cannot lift water to great heights.

Capillarity, the rise of water in very narrow tubes, makes a small additional contribution.

Transpiration

Transpiration is the loss of water vapour from the aerial parts of a plant, mainly through the stomata of the leaves. Water evaporates from the moist surfaces of the mesophyll cells into the air spaces and diffuses out through the stomata. A small amount passes through the cuticle (cuticular transpiration), and a little through lenticels on stems. Transpiration is useful, because it provides the pull that carries mineral salts up, cools the leaf in the tropical sun and supplies water for photosynthesis, but it is also a danger in dry weather because it may lead to wilting.

The rate of transpiration is increased by high temperature, low humidity, wind and bright light (which opens stomata) and decreased by the opposite conditions. A **potometer** is used to measure water uptake, which approximates the rate of transpiration. Plants of dry regions, called **xerophytes**, reduce water loss through a thick waxy cuticle, small or needle-like leaves, sunken stomata, hairs and, in cacti, storage of water in fleshy stems.

Guard cells and stomata

Each stoma is a pore bounded by two kidney-shaped **guard cells**. The inner wall of each guard cell is thicker than the outer. In the light, the guard cells accumulate solutes and take in water by osmosis; they become turgid and bow outward, so the pore opens. In darkness, or when the plant is short of water, they lose water, become flaccid and close the pore. Most stomata are on the lower surface of the leaf, where it is cooler, limiting water loss.

WORKED EXAMPLE 4.2

A potometer shows that a leafy shoot takes up water so that an air bubble moves 15 cm in 10 minutes along a capillary tube of cross-sectional area 0.02 cm². Calculate the volume of water absorbed per hour.

Solution: Volume in 10 minutes = 15 × 0.02 = 0.30 cm³. In 60 minutes the volume is 0.30 × 6 = 1.8 cm³ per hour.

Translocation in the phloem

Translocation is the movement of organic solutes through the phloem. The phloem contains **sieve tubes**, whose cell walls are perforated by sieve plates and which are living cells, accompanied by **companion cells** that supply the

energy and control their function. Sucrose is made in the leaves, loaded actively into the sieve tubes, and carried both upward and downward to growing tips, fruits, seeds and storage organs such as the yam tuber and cassava root. It can move in either direction, unlike water in the xylem, and it requires energy. Ringing the stem by removing a band of bark which contains phloem causes the stem to swell above the ring because sugars cannot pass downward, while the roots below eventually die of starvation.

4.5 Gaseous exchange

Gaseous exchange is the intake of oxygen and the release of carbon dioxide across a respiratory surface. Efficient surfaces share certain features: they are thin, moist, richly supplied with blood, large in area and, in animals, well ventilated so that a diffusion gradient is maintained.

Mammals: the lungs

Air enters through the nose, where it is warmed, moistened and filtered by hairs and mucus, passes through the pharynx, the larynx (voice box) and into the **trachea**, a tube kept open by C-shaped rings of cartilage. The trachea divides into two **bronchi**, one to each lung, and these divide repeatedly into **bronchioles** ending in clusters of tiny air sacs, the **alveoli**. An adult has some 300 million alveoli with a total surface area of roughly 70 m². Each is surrounded by a network of capillaries; oxygen dissolves in the moisture lining the alveolus and diffuses across the single-cell walls of alveolus and capillary into the blood, while carbon dioxide diffuses the opposite way.

Breathing is brought about by the diaphragm and the intercostal muscles.

Event	Inspiration (breathing in)	Expiration (breathing out)
Diaphragm	Contracts and flattens	Relaxes and becomes dome-shaped
External intercostal muscles	Contract, raising the ribs up and out	Relax, ribs fall down and in
Volume of thorax	Increases	Decreases
Pressure in lungs	Falls below atmospheric	Rises above atmospheric
Air movement	Into the lungs	Out of the lungs

Inhaled air has about 21 per cent oxygen, 0.04 per cent carbon dioxide and variable water vapour; exhaled air has roughly 16 per cent oxygen and 4 per cent carbon dioxide, with more water vapour, and is warmer. Carbon dioxide turns lime water milky, which is the basis of a simple laboratory demonstration. About 70 per cent of carbon dioxide is carried in the plasma as hydrogencarbonate ions, with the remainder dissolved or combined with haemoglobin. The respiratory rate is regulated by the brain, in response to the level of carbon dioxide in the blood, not oxygen.

The volume of air moved with each normal breath, the **tidal volume**, is about 500 cm³. The maximum volume that can be breathed out after the deepest possible breath is the **vital capacity**, around 4,500 cm³ in a healthy adult, and a volume of about 1,200 cm³, the residual volume, always stays in the lungs.

WORKED EXAMPLE 4.3

A student breathes 16 times per minute with a tidal volume of 500 cm^3. Calculate the volume of air that passes through his lungs in one hour, in litres.

Solution: Per minute = 16 × 500 = 8,000 cm^3. Per hour = 8,000 × 60 = 480,000 cm^3 = 480 litres.

Other organisms

- **Single-celled organisms** such as *Amoeba* exchange gases over the entire cell surface by diffusion.
- **Earthworms** breathe through the moist skin, which is supplied with capillaries.
- **Insects** have a system of branching air tubes, the **tracheae**, opening to the outside through small pores called **spiracles** along the thorax and abdomen. The tracheae carry air directly to the tissues, so insect blood plays no part in transporting oxygen.
- **Fishes** use **gills**, located behind the head under the operculum. Each gill has bony gill arches with red feathery **gill filaments** whose thin plates are packed with capillaries. Water taken in through the mouth passes over the gills and out under the operculum, and blood flows through the gill plates in a direction opposite to the water (counter-current flow), so that oxygen continues to diffuse into the blood along the entire length of the filament.
- **Amphibians** such as the frog use the skin, the lining of the mouth and the lungs; tadpoles use gills.
- **Plants** exchange gases through stomata and lenticels, and through air spaces in the leaf and in the roots. In the light, a plant takes in carbon

dioxide and gives out oxygen in photosynthesis; at the same time, it respires, taking in oxygen and releasing carbon dioxide.

Respiratory disorders

Smoking tobacco causes chronic bronchitis, emphysema and cancer of the lung, partly because tar paralyses and destroys the cilia that sweep dust out of the airway. Asthma is a narrowing of the bronchioles in response to an allergen. Tuberculosis is a bacterial infection of the lungs (Chapter 10), and pneumonia is an infection of the alveoli.

WORKED EXAMPLE 4.4

Explain, with reference to the structure of the alveolus, why smokers become short of breath when they exercise.

Solution: Tar and chemicals in smoke irritate the airways, causing excess mucus and damaging the cilia, so the air passages narrow and are partly blocked. In emphysema the walls of alveoli break down, reducing the surface area available for diffusion. With less area for gas exchange, less oxygen enters the blood, and the muscles cannot obtain enough for respiration at times of high demand, so the person breathes rapidly and feels breathless.

Exercise 4.1

1. Name the blood vessel that carries oxygenated blood from the lungs to the heart.
2. Why is the wall of the left ventricle thicker than that of the right?
3. A woman has blood group B. To which groups can she safely donate red cells, and from which can she receive?

4. A heart beats 68 times per minute, each beat expelling 70 cm^3 of blood. Calculate the cardiac output in litres per minute.
5. State two ways in which a red blood cell is adapted for its function.
6. Name the tissue in a plant that transports sucrose, and give two ways in which it differs from xylem.
7. List four factors that increase the rate of transpiration.
8. Calculate the surface-area-to-volume ratio of a cube with sides of 2 cm.
9. Explain how the guard cells open a stoma in the morning.
10. Describe the movement of the diaphragm and the ribs during inspiration.
11. In a potometer experiment, a bubble moves 24 cm in 12 minutes in a tube of cross-sectional area 0.05 cm^2. Calculate the rate of water uptake in cm^3 per minute.
12. Explain why fish die when taken out of water, even though air contains more oxygen than water.

Answers to Exercise 4.1

1. The pulmonary vein.
2. It must pump blood to the whole body at high pressure, whereas the right ventricle pumps only to the lungs.
3. Donate to B and AB; receive from B and O (B has anti-A antibodies, so it cannot receive A or AB).
4. $68 \times 70 = 4{,}760$ cm^3/min = 4.76 litres per minute.
5. Biconcave shape gives large surface area; no nucleus leaves room for more haemoglobin; haemoglobin binds oxygen.

6. Phloem. Phloem tubes are living and transport sugars in both directions, using energy; xylem vessels are dead, lignified and carry water and minerals upward only.
7. High temperature, low humidity, wind, bright light, (and a plentiful water supply); any four.
8. Surface area = $6 \times 2^2 = 24$ cm^2; volume = $2^3 = 8$ cm^3; ratio = 24/8 = 3 : 1.
9. Light causes accumulation of solutes in the guard cells; water enters by osmosis; they become turgid and, because of thicker inner walls, curve apart and open the pore.
10. The diaphragm contracts and flattens while the ribs move up and out, so the volume of the chest increases and the pressure falls.
11. Volume = $24 \times 0.05 = 1.2$ cm^3 in 12 minutes, so rate = $1.2 \div 12 = 0.1$ cm^3 per minute.
12. Without water support the gill filaments collapse and stick together, so the surface area for gas exchange is greatly reduced and the gills dry out; oxygen cannot diffuse in quickly enough.

Chapter summary

- Large organisms need transport systems because their surface-area-to-volume ratio is small.
- Blood consists of plasma, red cells (oxygen transport), white cells (defence) and platelets (clotting); ABO groups depend on antigens on red cells and antibodies in plasma, with O the universal donor and AB the universal recipient.
- The heart has four chambers; the double circulation sends deoxygenated blood to the lungs and oxygenated blood to the body; cardiac output equals heart rate times stroke volume.

- Arteries, veins and capillaries are adapted to pressure, return flow and exchange respectively.
- Water rises in the xylem by transpiration pull, cohesion and adhesion, and root pressure; sugars move in the phloem by translocation.
- Stomata open when guard cells become turgid in the light.
- Gaseous exchange surfaces are thin, moist, large and well supplied with blood; lungs, gills, tracheae and skin are the main examples.
- Inspiration involves contraction of the diaphragm and external intercostal muscles, increasing chest volume and lowering pressure.

Chapter 5: Respiration, Excretion and Homeostasis

Every living cell needs a continuous supply of energy to grow, move, divide, repair itself and carry out active transport. That energy is released from food molecules by the chemical process of respiration. At the same time, metabolism produces waste substances such as carbon dioxide, urea and excess water, which must be removed before they poison the body. This chapter explains how energy is released in cells, how the waste products of metabolism are excreted by the kidneys, lungs, skin and liver, and how the body keeps its internal conditions steady, a function called homeostasis.

Candidates should be able to write and interpret the equations of aerobic and anaerobic respiration, compare the two, label the nephron, explain what happens to blood sugar after a meal, and describe how the body responds to heat and cold. These are the questions treated here.

5.1 Respiration and energy

Respiration is the controlled release of energy from food in living cells. It is different from breathing, which is the physical movement of air in and out of the lungs. The energy released is captured in the molecule **adenosine triphosphate (ATP)**, the universal energy currency of the cell. ATP is made by adding a phosphate group to adenosine diphosphate (ADP), and when the terminal phosphate is removed again by hydrolysis energy is released for cell activities. Most of the energy not stored in ATP is lost as heat, which is why a body that is exercising becomes warm.

Glucose is the usual respiratory substrate. When it is exhausted, glycogen, fats and, in starvation, proteins can be used. The **respiratory quotient** (RQ) is the ratio of the volume of carbon dioxide produced to the volume of oxygen consumed. For glucose it is 1.0, for fats about 0.7 and for proteins about 0.9.

Aerobic respiration

Aerobic respiration uses oxygen and breaks glucose down completely to carbon dioxide and water:

$C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O$ + energy (about 2,880 kJ per mole of glucose; 38 ATP in textbook accounts, usually fewer in practice)

It occurs in two stages. **Glycolysis** takes place in the cytoplasm: the six-carbon glucose is split into two molecules of the three-carbon pyruvic acid, with a small gain of ATP, and this stage needs no oxygen. In the presence of oxygen, pyruvic acid enters the **mitochondria**, where the Krebs cycle and the electron transport chain, the stages involving oxygen, complete its oxidation, releasing carbon dioxide and water and most of the ATP. Mitochondria are called the "powerhouses" of the cell, and active cells such as those of muscle, of the sperm tail and of the kidney tubule have large numbers of them.

Anaerobic respiration

When oxygen is absent or inadequate, glucose is only partly broken down in the cytoplasm, and much less energy is released (about 2 ATP molecules per glucose). Two types are important:

- **Alcoholic fermentation** in yeast and some plant tissues: $C_6H_{12}O_6 \rightarrow 2C_2H_5OH + 2CO_2$ + energy (about 210 kJ per mole). The carbon dioxide makes bread dough rise, and the ethanol is the basis of brewing, as in the tapping of palm wine, in which yeast ferments the sugary sap.

- **Lactic acid fermentation** in animal muscle during vigorous exercise: $C_6H_{12}O_6 \rightarrow 2C_3H_6O_3$ + energy (about 150 kJ per mole). The lactic acid builds up and causes muscle fatigue and cramp. After exercise the accumulated lactic acid is oxidised using extra oxygen, and the volume of oxygen needed for this is the **oxygen debt**, which explains why one pants for some time after a hard sprint. Some bacteria, such as those in yoghurt and in the fermentation of ogi and fura da nono, also use lactic acid fermentation.

Feature	Aerobic	Anaerobic
Oxygen	Required	Not required
Site	Cytoplasm and mitochondria	Cytoplasm only
Breakdown of glucose	Complete	Incomplete
End products	CO_2 and water	Ethanol and CO_2, or lactic acid
Energy yield	High	Low

Key point. Anaerobic respiration yields only about 5 to 8 per cent of the energy per glucose molecule that aerobic respiration yields; energy remains locked in the ethanol or the lactic acid. That is why yeast must consume so much sugar in order to grow when it is without air.

Experiments on respiration

Respiration produces carbon dioxide, heat and uses oxygen, and each can be demonstrated. Carbon dioxide given off by germinating seeds or a mouse in a closed jar turns lime water milky or turns bicarbonate indicator from red to

yellow. Heat is shown by germinating seeds in a vacuum flask with a thermometer, the temperature of which rises compared with a control containing boiled, killed seeds. Oxygen uptake is shown with a respirometer, in which carbon dioxide is absorbed by potassium hydroxide so that the volume of the apparatus decreases as oxygen is used; the movement of a coloured liquid in a capillary tube indicates the volume of oxygen consumed. Fermentation by yeast in glucose solution under a layer of oil is shown by gas bubbles passing into lime water.

WORKED EXAMPLE 5.1

In a respirometer, germinating beans consume oxygen so that a drop of coloured liquid moves 4.5 cm in 30 minutes along a capillary tube of cross-sectional area 0.04 cm². Calculate the rate of oxygen uptake in cm³ per hour.

Solution: Volume of oxygen used in 30 minutes = 4.5 × 0.04 = 0.18 cm³. In one hour the volume is 0.18 × 2 = 0.36 cm³ per hour.

WORKED EXAMPLE 5.2

A sample of yeast ferments 18 g of glucose (relative molecular mass 180) according to $C_6H_{12}O_6 \rightarrow 2C_2H_5OH + 2CO_2$. Calculate the mass of ethanol formed (C = 12, H = 1, O = 16).

Solution: Moles of glucose = 18 ÷ 180 = 0.1 mol. The equation gives 2 mol of ethanol per mole of glucose, so ethanol = 0.2 mol. Molar mass of C_2H_5OH = 24 + 6 + 16 = 46 g/mol. Mass = 0.2 × 46 = 9.2 g.

5.2 Excretion

Excretion is the removal from the body of the waste products of metabolism, and of substances in excess of requirements. It must be separated from egestion, the removal of undigested food, and from secretion, in which a useful product is released by a gland. The main excretory products and the organs involved are:

Organ	Excretory product
Kidneys	Urea, excess water and salts, uric acid, creatinine
Lungs	Carbon dioxide and water vapour
Skin	Sweat: water, salts and a little urea
Liver	Bile pigments (from the breakdown of haemoglobin), which leave through the gut; urea is made here

The urinary system

The human urinary system consists of a pair of bean-shaped **kidneys** at the back of the abdomen, one on each side of the backbone, a **ureter** leaving each kidney, a muscular **bladder** where urine is stored, and a **urethra** through which urine is passed to the exterior. Blood reaches each kidney by the **renal artery** and leaves through the **renal vein**. Blood in the renal vein has less urea, less water and fewer salts than that in the renal artery, and it has less oxygen and more carbon dioxide because the kidney tissue is itself respiring.

A longitudinal section of the kidney shows an outer **cortex**, an inner **medulla** composed of pyramids, and a central cavity, the **pelvis**, which leads into the ureter. Each kidney contains about one million tiny filtering tubes called **nephrons**.

The nephron

A nephron begins with a cup-shaped **Bowman's capsule** in the cortex, enclosing a knot of capillaries called the **glomerulus**. The capsule, together with the glomerulus, is called the Malpighian body. From the capsule the tubule runs as the proximal convoluted tubule, then the loop of Henle dipping into the medulla, then the distal convoluted tubule, and finally joins a collecting duct that carries urine to the pelvis. Urine formation involves three processes.

1. **Ultrafiltration** in the glomerulus. The afferent vessel carrying blood into the glomerulus is wider than the efferent vessel taking blood out, so the pressure in the glomerulus is high. Water, glucose, amino acids, urea and salts are forced through the thin capillary wall and the capsule into the tubule, forming the glomerular filtrate. Blood cells and large proteins are too big to pass and remain in the blood. About 180 litres of filtrate are formed per day in an adult.
2. **Selective reabsorption**. In the proximal convoluted tubule all the glucose and amino acids, and most of the salts and water, are reabsorbed into the capillaries around the tubule by active transport and diffusion. More water is reabsorbed in the loop of Henle and the collecting duct.
3. **Tubular secretion**. Certain substances, for example some drugs, potassium and hydrogen ions, are actively secreted from the blood into the distal tubule.

The fluid that remains, containing water, urea, uric acid, salts and pigments, is **urine**, about 1.5 litres per day. Healthy urine contains no glucose and no protein. The presence of glucose indicates diabetes mellitus, and of protein suggests kidney damage.

WORKED EXAMPLE 5.3

An adult forms 180 litres of glomerular filtrate in a day and passes 1.5 litres of urine. What percentage of the filtrate is reabsorbed? How much water is reabsorbed each hour (assuming that the urine is nearly all water)?

Solution: Water reabsorbed = 180 − 1.5 = 178.5 litres. Percentage = (178.5 ÷ 180) × 100 = 99.2 per cent. Per hour: 178.5 ÷ 24 = 7.44 litres.

Osmoregulation and the role of ADH

The concentration of the blood is monitored by the hypothalamus of the brain. When the body is short of water (after sweating or too little drinking), the blood becomes more concentrated and the pituitary gland releases more **antidiuretic hormone (ADH)**. ADH makes the walls of the distal tubules and collecting ducts more permeable to water, so more water is reabsorbed and a small volume of concentrated urine is produced. When one drinks a lot of water, the blood is diluted, less ADH is released, and a large volume of dilute urine results. This is an example of negative feedback. A lack of ADH causes **diabetes insipidus**, in which very large volumes of dilute urine are produced. Alcohol suppresses ADH, which explains the frequent urination of people who drink beer.

Kidney disorders

Kidney failure can be treated by **dialysis**, in which the patient's blood passes through a machine with partially permeable tubes immersed in a fluid that removes urea and excess salts but keeps the blood proteins and cells; or by **kidney transplant**, which may be rejected by the immune system unless

tissues are closely matched and immunosuppressant drugs are used. **Kidney stones** are hard deposits of calcium salts or uric acid, and **nephritis** is inflammation of the nephrons.

5.3 The skin and the liver as excretory organs

The skin has two layers. The outer **epidermis** has a surface of dead keratinised cells and a deeper germinative layer producing new cells and the pigment melanin. The inner **dermis** contains blood capillaries, sweat glands, sebaceous glands which secrete oily sebum, hair follicles with the erector muscles, sensory receptors for touch, pain, heat and cold, and fat in the layer beneath. The skin protects against germs, water loss and harmful ultraviolet light, produces vitamin D in sunlight, excretes sweat and, above all, regulates body temperature.

The liver is the largest gland of the body, weighing about 1.5 kg. Beyond making bile and storing glycogen, as noted in Chapter 3, it is the site where excess amino acids are deaminated and the toxic ammonia is converted to urea by a series of reactions called the ornithine cycle. It breaks down worn red blood cells, forming bile pigments, and removes hormones and poisons. It stores iron and vitamins, and makes plasma proteins and cholesterol.

5.4 Homeostasis

Homeostasis is the maintenance of a relatively constant internal environment, the blood and tissue fluid, in the face of changes outside. It depends on **negative feedback**: a change in a factor is detected by receptors, and a response is triggered which reverses the change and returns the factor to its

normal level. The factors regulated include body temperature (about 37 °C), blood glucose concentration, water content, blood pH and the level of carbon dioxide.

Temperature regulation

Humans are **endotherms** (homeotherms), keeping a nearly constant body temperature by internal means; reptiles, amphibians and fishes are **ectotherms** (poikilotherms), whose temperature follows that of their surroundings. The temperature centre is the hypothalamus, which monitors the temperature of blood passing through it and receives information from temperature receptors in the skin.

If the body is too hot	If the body is too cold
Sweat glands produce sweat, whose evaporation takes up latent heat	Sweating stops
Vasodilation: skin arterioles widen, more blood flows near the surface and heat is lost by radiation	Vasoconstriction: skin arterioles narrow, so less heat is lost
Hair erector muscles relax, hairs lie flat	Hairs raised (goose pimples) to trap an insulating layer of air
Metabolic rate falls	Shivering and a rise in metabolic rate produce heat; adrenaline raises respiration

A fever is a rise in body temperature, often caused by infection such as malaria. Heat stroke occurs when the heat-loss mechanisms fail in very hot conditions.

Control of blood glucose

The pancreas contains groups of cells called the **islets of Langerhans**, which monitor the glucose concentration of the blood passing through. The normal fasting level is about 80 to 100 mg per 100 cm^3 of blood. After a meal the glucose level rises; the beta cells release **insulin**, which makes the liver and muscles convert glucose to glycogen, and increases the uptake and use of glucose by cells, so the blood glucose falls. Between meals or in exercise the level falls; the alpha cells release **glucagon**, which causes the liver to convert glycogen back to glucose, restoring the level. Adrenaline also raises blood glucose in emergencies.

Diabetes mellitus results when the pancreas produces too little insulin or the body cells do not respond to it. Blood glucose rises so high that the kidney cannot reabsorb it all, and glucose appears in the urine. Symptoms are thirst, frequent urination, tiredness and weight loss. Type 1 diabetes is treated by injections of insulin and a controlled diet; type 2 diabetes is often controlled by diet, exercise and tablets.

> **Key point.** Insulin lowers blood sugar; glucagon raises it. Insulin is secreted by the beta cells of the islets of Langerhans in the pancreas and works by promoting glycogen formation in the liver. Memorise this opposition because it is the textbook case of negative feedback.

Regulation of other factors

Blood pH is kept near 7.4 by the buffering action of plasma proteins and hydrogencarbonate and by the removal of carbon dioxide at the lungs. A rise in carbon dioxide makes the blood more acidic and is detected in the brain,

which increases the breathing rate. Water and salt balance is controlled by the kidney under the influence of ADH and the hormone aldosterone, secreted by the adrenal cortex, which increases sodium reabsorption.

WORKED EXAMPLE 5.4

Explain what happens in the body of a footballer in Lagos who, after drinking a large bottle of water, has a very pale urine of large volume.

Solution: The extra water dilutes the blood plasma, and the hypothalamus detects the reduced concentration. The pituitary gland releases less ADH, so the distal tubules and collecting ducts become less permeable to water, little water is reabsorbed and a large volume of dilute, pale urine passes out. This restores the water content of the blood to normal, a case of negative feedback.

Exercise 5.1

1. Write the word equation for anaerobic respiration in a muscle.
2. Where in the cell does glycolysis occur, and does it need oxygen?
3. Why does an athlete continue to breathe heavily after finishing a 100 m race?
4. State two differences between excretion and egestion.
5. In which part of the nephron does ultrafiltration take place, and why is the filtrate free of red blood cells?
6. A woman forms 170 litres of filtrate in a day and passes 1.7 litres of urine. Calculate the percentage of filtrate reabsorbed.
7. Name the hormone that increases the permeability of the collecting duct, and say where it is secreted.

8. Name two substances that are normally absent from urine but present in a diabetic patient and one in a patient with kidney damage.
9. Explain how vasoconstriction helps in cold weather.
10. Calculate the mass of carbon dioxide produced when 36 g of glucose ferment to ethanol (C = 12, H = 1, O = 16): $C_6H_{12}O_6 \rightarrow 2C_2H_5OH + 2CO_2$.
11. Describe how the body reduces the blood glucose concentration after a heavy meal.
12. A respirometer drop moves 3 cm in 15 minutes in a capillary of cross-sectional area 0.02 cm². Calculate the oxygen uptake per hour.

Answers to Exercise 5.1

1. Glucose → lactic acid + energy.
2. In the cytoplasm; it does not need oxygen.
3. To repay the oxygen debt, i.e. to oxidise the lactic acid formed anaerobically in the muscles.
4. Excretion is the removal of metabolic waste made in cells (such as urea); egestion is the removal of undigested food that never entered the cells. Excretion is through kidneys, lungs and skin; egestion is through the anus.
5. In the glomerulus and Bowman's capsule; red cells and large protein molecules are too big to pass through the capillary walls and the capsule.
6. Reabsorbed = 170 − 1.7 = 168.3 litres; (168.3 ÷ 170) × 100 = 99 per cent.
7. ADH (antidiuretic hormone), released by the posterior pituitary gland (made in the hypothalamus).
8. Glucose (and ketones) in diabetes; protein in kidney damage.
9. The skin arterioles narrow, so less blood flows near the surface and less heat is lost by radiation.

10. 36 g = 0.2 mol glucose, giving 0.4 mol CO_2; molar mass of CO_2 = 44; mass = 0.4 × 44 = 17.6 g.
11. The pancreas secretes insulin; the liver and muscles convert glucose to glycogen, and cells take up and use more glucose, so blood glucose falls.
12. Volume in 15 minutes = 3 × 0.02 = 0.06 cm^3; per hour = 0.06 × 4 = 0.24 cm^3.

Chapter summary

- Respiration releases energy from food and stores it in ATP; aerobic respiration is complete and releases much more energy than anaerobic respiration.
- Aerobic: glucose + oxygen → carbon dioxide + water + energy. Anaerobic: glucose → ethanol + CO_2 in yeast; glucose → lactic acid in muscle.
- Excretion removes metabolic wastes through the kidneys, lungs, skin and liver.
- The nephron forms urine by ultrafiltration, selective reabsorption and tubular secretion; ADH controls water reabsorption.
- Homeostasis maintains temperature, glucose, water and pH by negative feedback.
- Insulin lowers and glucagon raises blood glucose; diabetes mellitus is a failure of this control.
- The skin regulates temperature through sweating, vasodilation and vasoconstriction, and hair erection.

Chapter 6: Reproduction, Heredity and Variation

Reproduction is the production of new individuals of the same kind, and through it life continues from one generation to the next. Heredity is the passing of characteristics from parents to offspring, and variation is the differences that exist between individuals of the same species. Together they form the foundation of genetics and of the theory of evolution treated in Chapter 7. In this chapter we study asexual and sexual reproduction in plants and animals, human reproduction and the menstrual cycle, and then the laws of inheritance discovered by Gregor Mendel, with the calculations that the UTME regularly sets.

6.1 Asexual reproduction

In **asexual reproduction** a single parent produces offspring without gametes or fertilisation. The offspring are genetically identical to the parent and to one another, and together are called a **clone**, because all of the cell divisions involved are mitotic. It is rapid and needs no partner, but it gives no variation, so a disease or a change in conditions can destroy a whole population. The main forms are:

- **Binary fission**: the parent divides into two equal daughters, as in *Amoeba*, *Paramecium* and bacteria.
- **Budding**: a small outgrowth develops on the parent and breaks away, as in yeast and *Hydra*.
- **Spore formation**: tiny light spores are released from sporangia, as in *Rhizopus* (bread mould), mushrooms and ferns, and germinate where

conditions are right.

- **Fragmentation and regeneration**: a body breaks into pieces, each growing into a new individual, as in the flatworm *Planaria* and some algae.
- **Vegetative propagation** in flowering plants: new plants grow from parts of the parent. Examples are **tubers** (yam, Irish potato), **rhizomes** (ginger, banana, spear grass), **bulbs** (onion), **corms** (cocoyam), **stolons or runners** (strawberry, *Cynodon*), **suckers** (banana, plantain) and the leaf buds on the leaf margins of the leaf-of-life plant (*Bryophyllum*). Farmers also use stem cuttings (cassava, sugarcane), grafting and budding to propagate improved varieties of citrus, mango and cocoa.

6.2 Sexual reproduction

In **sexual reproduction**, two specialised sex cells, the **gametes**, fuse in **fertilisation** to form a **zygote**. The gametes are made by meiosis (Chapter 2 treated mitosis; meiosis is described below), so each has half the normal number of chromosomes (**haploid**, n), and the zygote has the full (**diploid**, 2n) number. The male gamete is small and mobile (sperm, or the pollen nucleus), and the female gamete is larger and stationary (egg or ovum). Sexual reproduction creates variation, since offspring receive a new combination of genes from two parents, and so it provides the raw material for natural selection.

Meiosis

Meiosis is the type of cell division in the gonads (testes and ovaries) and in the anthers and ovules of flowers, in which a diploid cell divides twice to give four haploid cells. In the first division homologous chromosomes pair up, and

the pairs, not the single chromatids, are separated. During pairing, segments may be exchanged between chromatids in a process called **crossing over**. In the second division, the chromatids separate as in mitosis. Human body cells have 46 chromosomes (23 pairs) and human gametes 23. The significance of meiosis is that the number of chromosomes is kept constant from generation to generation, and that crossing over and the random assortment of chromosomes give genetic variation.

Mitosis	Meiosis
One division, two daughter cells	Two divisions, four daughter cells
Chromosome number unchanged (2n to 2n)	Chromosome number halved (2n to n)
Daughters identical to the parent cell	Daughters genetically different
No pairing or crossing over	Homologous pairing and crossing over
Growth, repair, asexual reproduction	Formation of gametes or spores

6.3 Reproduction in flowering plants

The flower is the reproductive organ of an angiosperm. A typical flower has four whorls, from the outside inwards: the **calyx** of green sepals protecting the bud; the **corolla** of petals, usually coloured to attract insects; the **androecium**, the male part, made up of **stamens** each having an **anther** on a filament; and the **gynoecium** (carpels or pistil), the female part, with the **stigma**, the **style** and the **ovary** containing one or more **ovules**. A flower with both stamens and carpels is bisexual, as in hibiscus, while maize bears separate male flowers (tassel) and female flowers (cob) on the same plant, and pawpaw usually has male and female flowers on separate plants.

Pollination

Pollination is the transfer of pollen grains from an anther to a stigma. It is **self-pollination** when the pollen goes to the stigma of the same flower or of another flower on the same plant, and **cross-pollination** when it goes to a flower on a different plant of the same species. Cross-pollination produces more vigorous and varied offspring but depends on an agent.

Feature	Insect-pollinated flower	Wind-pollinated flower
Petals	Large, brightly coloured, scented	Small, dull or absent
Nectar	Present	Absent
Pollen	Sticky, spiny, relatively few	Smooth, light, very abundant
Anthers	Inside the flower	Hanging outside, loosely attached
Stigma	Small, sticky, inside	Large, feathery, outside
Examples	Hibiscus, cashew, bean	Maize, guinea corn, grasses

Fertilisation, seed and fruit

A pollen grain which lands on a compatible stigma develops a **pollen tube** that grows down through the style, carrying two male nuclei, to the ovule and enters it through a small opening called the micropyle. One male nucleus fuses with the egg nucleus to form the diploid zygote, and the other fuses with the two polar nuclei to form a triploid endosperm nucleus. This is **double fertilisation**, which is a feature of the flowering plants. The zygote develops into the **embryo**, with a radicle (young root) and a plumule (young shoot), the endosperm becomes a food store, the ovule becomes the **seed** and the ovary becomes the **fruit**. Fruit may be fleshy (mango, tomato, orange) or dry (groundnut pod, maize grain).

Seeds are **dispersed** so that the seedlings do not compete with the parent. Wind disperses light or winged seeds (the silk cotton and *Tridax*, with parachutes of hairs); animals carry hooked fruits such as *Bidens* or eat fleshy fruits and pass out the seeds; water disperses coconut; and explosive dehiscence of the dry pods of the flamboyant, *Crotalaria* and castor oil scatters the seeds.

Germination needs water, oxygen and a suitable temperature; light is not necessary for most seeds. Germination is **epigeal** if the cotyledons are carried above the ground (bean, castor oil, pawpaw), and **hypogeal** if the cotyledons stay below the soil (maize, groundnut, cowpea, mango).

6.4 Reproduction in animals

Most animals reproduce sexually. Fertilisation may be **external**, with eggs and sperm released into water, as in most fishes and amphibians such as the frog, in which a large number of eggs is laid because many are lost; or **internal**, as in reptiles, birds and mammals, where eggs are fewer and better protected. Insects such as the housefly, mosquito and butterfly undergo **metamorphosis**: in *complete* metamorphosis the egg gives a larva, then a pupa, and then an adult (egg, larva, pupa, imago); in *incomplete* metamorphosis, as in the grasshopper and cockroach, the egg hatches into a nymph that looks like a small adult and grows by moulting. Frogs go through a metamorphosis from tadpole to adult.

6.5 Human reproduction

The male system

The two **testes** are held in the scrotum outside the abdomen, where the temperature is a few degrees below body temperature, which is necessary for sperm production. Inside them the seminiferous tubules produce sperm by meiosis, and interstitial cells secrete the male hormone **testosterone**. Sperm pass to the epididymis, where they mature and are stored, and then along the **sperm duct (vas deferens)** to the urethra. The seminal vesicles, the prostate gland and Cowper's gland add fluids that provide nutrients, an alkaline medium and lubrication; the mixture is semen. The urethra passes through the penis. A sperm has a head containing the nucleus and a cap of enzymes (the acrosome), a middle piece packed with mitochondria to provide energy, and a long tail for swimming.

The female system

The two **ovaries** in the lower abdomen produce ova and the hormones **oestrogen** and **progesterone**. An ovum is released at **ovulation** into the funnel of the **oviduct (Fallopian tube)**, where fertilisation usually occurs. The fertilised egg moves to the **uterus (womb)**, a muscular organ with a lining called the endometrium, where it implants. The uterus opens to the outside through the cervix and the **vagina**.

The menstrual cycle

From puberty to the menopause, the female reproductive system passes through a cycle averaging 28 days. Day 1 is the first day of menstruation, the loss of the uterine lining with blood, which lasts about 4 to 5 days. Follicle-stimulating hormone (FSH) from the pituitary causes a follicle to ripen in the

ovary and secrete oestrogen, and oestrogen causes repair and thickening of the uterine lining. A surge of luteinising hormone (LH) triggers **ovulation** at about day 14. The empty follicle becomes the **corpus luteum**, which secretes progesterone to maintain the lining. If the ovum is not fertilised, the corpus luteum degenerates, hormone levels fall, and the lining breaks down in menstruation. If pregnancy occurs, the corpus luteum persists and progesterone keeps the lining, so menstruation does not occur.

WORKED EXAMPLE 6.1

A woman has a regular menstrual cycle of 28 days, the first day of her last period being 3 March. On what date is ovulation most probable, and over which dates is she most likely to conceive? (Sperm live up to about 3 days, the ovum about one day.)

Solution: Ovulation occurs about 14 days from the start of the cycle, so it falls on day 14. Counting 3 March as day 1, day 14 is 16 March. The fertile period runs from about three days before ovulation to one day after, that is, 13 March to 17 March.

Fertilisation, pregnancy and birth

During copulation, semen is deposited in the vagina; sperm swim through the cervix and uterus into the oviduct, and one sperm penetrates the ovum. The nuclei fuse to give a diploid zygote, which divides repeatedly as it travels down the oviduct, and about seven days after fertilisation it sinks into the uterine lining (**implantation**). The **placenta**, formed from tissues of both embryo and mother, is joined to the embryo by the umbilical cord. Maternal and fetal blood do not mix, but oxygen, food, antibodies and some drugs pass from mother to child across the placenta by diffusion, and carbon dioxide and urea go the other way. The embryo is surrounded by the **amnion**, containing

amniotic fluid which cushions it against shock. The placenta also secretes progesterone. Pregnancy lasts about 40 weeks (nine months) and birth is brought about by contractions of the uterus under the influence of the hormone oxytocin. After birth the baby is fed with **breast milk**, which contains the right nutrients and antibodies, a first milk called colostrum being especially rich in them.

Twins are of two types: identical (monozygotic) twins come from one fertilised egg that splits, and so are always of the same sex and genetically identical; fraternal (dizygotic) twins come from two eggs fertilised by two sperm, and may be of different sexes. **Family planning** methods include barrier methods (condoms, which also reduce the risk of sexually transmitted infections), hormonal pills and injections, the intra-uterine device, the rhythm (calendar) method and sterilisation.

6.6 Heredity and variation

Variation is of two kinds. **Continuous variation** has a gradual range of values, for example height, mass and skin colour; it is controlled by many genes and is influenced by the environment. **Discontinuous variation** has distinct classes with no intermediates, for instance blood group, the ability to roll the tongue, sex and the sickle-cell trait; it is controlled by one or a few genes and is little affected by the environment.

The units of inheritance are **genes**, sections of DNA on the chromosomes. Different forms of the same gene are called **alleles**. The following terms are used in genetics.

- **Genotype**: the genetic make-up of an organism, e.g. Tt. **Phenotype**: the observable characteristic, e.g. tall.

- **Dominant** allele: the one expressed even when only one copy is present (written with a capital letter). **Recessive** allele: expressed only when both copies are recessive (small letter).
- **Homozygous** (pure-breeding): two identical alleles (TT or tt). **Heterozygous** (hybrid): two different alleles (Tt).
- **F_1**: the first filial generation; **F_2**: the offspring of the F_1.

Mendel's laws

Gregor Mendel, an Austrian monk, crossed varieties of garden pea in the 1860s. He concluded that characters are determined by pairs of factors (now genes), and the two factors separate during the formation of gametes, so each gamete carries only one. This is the **law of segregation**. When two characters are followed together, they are inherited independently of each other, the **law of independent assortment**.

Monohybrid inheritance

When a pure tall pea plant (TT) is crossed with a pure dwarf plant (tt), all the F_1 are Tt and tall. When the F_1 are crossed with each other, the gametes from each parent are T and t in equal numbers, and the F_2 are in the ratio 1 TT : 2 Tt : 1 tt, which is 3 tall : 1 dwarf. A **test cross**, in which an individual of dominant phenotype is crossed with a homozygous recessive (tt) to find out whether it is TT or Tt, gives all tall offspring if the individual is TT, and 1 tall : 1 dwarf if it is Tt.

WORKED EXAMPLE 6.2

In cattle, the hornless condition (H) is dominant over horned (h). A hornless bull mated with horned cows produces 12 calves, of which 5 are horned. What is the genotype of the bull, and how many hornless calves were born?

Solution: Horned calves are hh, so they received an h from the bull. The bull is hornless and carries h, so his genotype is Hh. The cross Hh × hh gives 1 Hh : 1 hh, or 1 hornless : 1 horned. Hornless calves = 12 − 5 = 7 (close to the expected 6 : 6).

Incomplete dominance and codominance

In **incomplete dominance** neither allele is fully dominant and the heterozygote is intermediate; red-flowered and white-flowered four-o'clock plants give pink F_1, and the F_2 ratio is 1 red : 2 pink : 1 white. In **codominance**, both alleles are fully expressed in the heterozygote; the AB blood group is the usual example.

Blood groups and the sickle-cell gene

The ABO group is controlled by three alleles: I^A and I^B, which are codominant, and i, which is recessive. Group A is I^AI^A or I^Ai; group B is I^BI^B or I^Bi; group AB is I^AI^B; group O is ii.

The gene for adult haemoglobin has the normal allele (Hb^A) and the sickle allele (Hb^S). A person with Hb^AHb^A is normal; Hb^SHb^S suffers sickle-cell anaemia, in which the red cells become sickle-shaped when oxygen is low, block capillaries and cause pain and anaemia; and Hb^AHb^S has the sickle-cell **trait** and is usually healthy, being also more resistant to malaria. This is why the gene remains common in West Africa, where malaria is frequent: the

heterozygote has a survival advantage. Two carriers (AS × AS) have a one-in-four chance in each pregnancy of having a child with sickle-cell anaemia (SS), a one-in-two chance of a carrier child, and a one-in-four chance of a normal child (AA). This is why genotype testing before marriage is advised.

WORKED EXAMPLE 6.3

A man of blood group A whose mother was group O marries a woman of group B whose father was group O. Find the possible blood groups of their children and the probability of each.

Solution: The man is group A with an O mother, so he received i from her: genotype I^Ai. The woman is group B with an O father: genotype I^Bi. The man's gametes are I^A and i; the woman's gametes are I^B and i. The four equally likely combinations are I^AI^B (group AB), I^Ai (A), I^Bi (B) and ii (O). Each of the groups A, B, AB and O therefore has a probability of 1/4.

Dihybrid inheritance

When two pairs of contrasting characters are followed, a cross between two double heterozygotes (AaBb × AaBb) gives a phenotypic ratio of 9 : 3 : 3 : 1 in the F_2. For example, if round (R) is dominant to wrinkled and yellow (Y) is dominant to green, the F_2 contains 9 round yellow : 3 round green : 3 wrinkled yellow : 1 wrinkled green. Out of 16 plants in the F_2, nine have at least one R and one Y allele.

WORKED EXAMPLE 6.4

A cross between two round-yellow pea plants, both RrYy, yields 960 offspring. Calculate the expected number of wrinkled green peas and of round green peas.

Solution: The ratio is 9 : 3 : 3 : 1, a total of 16 parts. Wrinkled green (rryy) is 1/16, so 960 ÷ 16 = 60. Round green (R_yy) is 3/16, so 3 × 60 = 180.

Sex determination and sex linkage

Of the 23 pairs of human chromosomes, 22 are **autosomes** and one pair is the **sex chromosomes**: XX in females and XY in males. Every ovum carries an X; half of the sperm carry X and half carry Y. The sex of the child is therefore decided by the father's sperm, and the probability of a boy or a girl is 1/2 each. Genes carried on the X chromosome show **sex-linked** inheritance. Because males have only one X, a recessive allele on it is always expressed in them. **Haemophilia** and **red-green colour blindness** are recessive and X-linked, and so are much commoner in men. A woman who is heterozygous (X^HX^h) is a carrier and normal; if she marries a normal man (X^HY), half her sons are expected to be affected and half her daughters are expected to be carriers. Affected men pass the allele to all of their daughters but none of their sons.

Mutation

A **mutation** is a sudden change in the genes or in the number or structure of chromosomes. Gene mutations, such as that giving sickle-cell haemoglobin, change the base sequence of DNA; chromosome mutations include Down's syndrome, in which a person has three copies of chromosome 21 (47 chromosomes). Mutations arise spontaneously and are increased by **mutagens**

like X-rays, gamma rays, ultraviolet light and some chemicals. Most are harmful, but occasionally one is advantageous, and mutation is the original source of all new alleles.

Exercise 6.1

1. State two differences between mitosis and meiosis.
2. Name the type of vegetative propagation used for (a) yam and (b) ginger.
3. Why are wind-pollinated flowers often without bright petals?
4. State the roles of the corpus luteum and of progesterone in the menstrual cycle.
5. Explain why the placenta is important to the unborn baby.
6. A cross between two heterozygous tall plants (Tt × Tt) produces 600 seedlings. How many are expected to be dwarf?
7. A man of blood group AB marries a woman of group O. What groups may their children have?
8. Two parents with normal haemoglobin each have a child with sickle-cell anaemia. What are the genotypes of the parents, and what is the chance that their next child will be a carrier?
9. A colour-blind man marries a woman who is homozygous for normal vision. What proportion of their sons and daughters is expected to be colour-blind, and what are the daughters' genotypes?
10. In a dihybrid cross of AaBb × AaBb, what fraction of the offspring is expected to be aabb?
11. A woman has a cycle of 30 days starting on 10 June. Estimate the day of ovulation (taken as 14 days before the next period).

12. Distinguish between genotype and phenotype, with an example.

Answers to Exercise 6.1

1. Mitosis: one division giving two identical diploid cells; meiosis: two divisions giving four genetically different haploid cells, with crossing over.
2. (a) Stem tuber; (b) rhizome.
3. They do not need to attract insects; large bright petals would block the free flow of wind-borne pollen.
4. The corpus luteum (the empty follicle) secretes progesterone, which maintains the thickened uterine lining for implantation; it degenerates if there is no pregnancy.
5. It allows oxygen, food and antibodies to pass to the fetus and carbon dioxide and urea to leave, without mixing maternal and fetal blood; it also secretes progesterone.
6. Dwarf (tt) is 1/4 of the offspring: 600 ÷ 4 = 150.
7. The man is I^AI^B, the woman ii; the children are I^Ai (A) or I^Bi (B), so group A or group B, in equal proportions.
8. Both parents are Hb^AHb^S (carriers); the chance of a carrier is 1/2.
9. The father is X^cY; the mother X^CX^C. Sons receive Y from the father and X^C from the mother: none colour-blind. All daughters are X^CX^c, carriers with normal vision.
10. 1/16.
11. The next period begins on 10 June + 30 days = 10 July; ovulation about 14 days earlier, on 26 June.
12. Genotype is the genetic make-up (e.g. Tt); phenotype is the visible character that results (e.g. tall).

Chapter summary

- Asexual reproduction involves one parent and gives clones; sexual reproduction involves the fusion of haploid gametes, producing variation.
- Meiosis halves the chromosome number and gives four different cells; mitosis keeps it constant.
- Flowers are pollinated by insects or wind; after double fertilisation the ovule becomes the seed and the ovary the fruit.
- The menstrual cycle is controlled by FSH, LH, oestrogen and progesterone, with ovulation at about day 14.
- The placenta exchanges materials between mother and fetus; pregnancy lasts about 40 weeks.
- Mendel's laws: segregation and independent assortment; monohybrid ratio 3 : 1; dihybrid ratio 9 : 3 : 3 : 1; test cross 1 : 1.
- Blood groups show multiple alleles and codominance; sickle-cell and haemophilia are classic examples of recessive and sex-linked inheritance.
- Mutations are sudden changes in genes or chromosomes and are the source of new alleles.

Chapter 7: Evolution and the History of Life

Chapter 6 showed how characters are inherited and how new alleles arise through mutation. This chapter asks what happens to those alleles over many generations. The answer is the theory of evolution: the idea that all living things share common ancestry and that populations change gradually through time. Evolution is the unifying principle of modern biology. It explains why the pentadactyl limb of a bat resembles the arm of a farmer, why bacteria become resistant to antibiotics, why the Nigerian rainforest and the Sahel contain such different communities of organisms, and why the fossil record shows a long procession of forms, many of them now extinct.

The UTME syllabus expects you to define evolution, outline the evidence for it, state the theories of Lamarck and Darwin and distinguish them, explain adaptation and natural selection, describe speciation, and give examples such as industrial melanism and antibiotic resistance. Questions are usually conceptual, so the emphasis below is on precise definitions and on being able to tell similar-sounding ideas apart.

7.1 What evolution means

Evolution is the gradual change in the inherited characteristics of populations of organisms from one generation to the next, leading over very long periods to the appearance of new species. Three points in this definition need emphasis. First, the unit that evolves is the *population*, not the individual. A single cheetah does not evolve during its lifetime; the frequency of particular alleles in the cheetah population shifts across generations. Second, only

inherited changes count. A bodybuilder who develops large muscles has not changed the genes he passes to his children. Third, the process is slow on a human time scale, although some cases, such as bacteria, can be observed within years.

Evolution is sometimes called **organic evolution** to distinguish it from the everyday use of the word for any gradual change, such as the evolution of fashion. Because all organisms are descended from earlier forms, classification (Chapter 1) is in effect a record of evolutionary relationship: organisms placed in the same genus are more closely related than those placed only in the same family.

> **Key point.** Evolution acts on populations, requires heritable variation, and takes place over generations. Changes acquired during an individual's lifetime are not evolution.

7.2 Evidence for evolution

No single observation proves evolution. The conclusion rests on several independent lines of evidence that point the same way.

Fossil records

A **fossil** is the preserved remains, impression or trace of an organism that lived in the geological past. Fossils form most readily when an organism is buried quickly in sediment such as mud or silt, so that its hard parts (shells, bones, teeth, wood) are protected from decay and are slowly replaced by minerals. Other fossils are moulds and casts, footprints, and whole insects trapped in amber. Rocks are laid down in layers called strata, with the oldest at the

bottom unless they have been disturbed. When fossils are examined layer by layer, a pattern emerges: the lowest layers contain simple organisms, and more complex forms appear in later layers. The fossil horse series, for example, shows a gradual increase in body size, a lengthening of the limbs, a reduction in the number of toes from four to one, and a change in the teeth for grazing on grass. *Archaeopteryx*, a fossil with both reptilian features (teeth, a long bony tail, clawed fingers) and bird features (feathers, wings), is a classic transitional form linking reptiles and birds.

The age of rocks and fossils is estimated by **radiometric dating**, which uses the steady decay of radioactive isotopes. Carbon-14, with a half-life of about 5,700 years, is used for remains younger than roughly 50,000 years, while isotopes such as uranium-238 are used for much older rocks.

Comparative anatomy

Comparing the structures of different animals reveals similarities that are best explained by common ancestry.

- **Homologous structures** have the same basic structure and embryonic origin but may have different functions. The forelimbs of a human, a bat, a whale and a horse all have the same plan of bones: one upper-arm bone (humerus), two lower-arm bones (radius and ulna), wrist bones (carpals), palm bones (metacarpals) and finger bones (phalanges). This is the **pentadactyl limb**. The same plan is modified for grasping, flying, swimming or running. Such modification of a common ancestral structure for different functions is called **adaptive radiation** or divergent evolution.
- **Analogous structures** have different basic structure and origin but similar function. The wing of a bird (a modified forelimb with feathers) and the wing of an insect (a fold of the exoskeleton) both serve for flight but are

not built on the same plan. Analogy arises through **convergent evolution**, in which unrelated organisms adapt to similar ways of life.

- **Vestigial structures** are reduced, functionless or nearly functionless remnants of organs that were fully functional in ancestors. Examples in humans include the appendix, the tail bone (coccyx) and the muscles that move the ears; in whales, the small pelvic bones; in snakes, the rudimentary hind-limb bones in pythons.

> **Key point.** Homologous means same origin, different function (evidence of common ancestry). Analogous means same function, different origin (evidence of convergent evolution).

Comparative embryology

The early embryos of fish, amphibians, reptiles, birds and mammals look strikingly alike. All possess gill slits and a tail at some stage. In fish the gill slits develop into functional gills, but in humans they are modified into structures of the ear and throat. Such resemblance in development suggests that vertebrates share a common ancestor.

Comparative biochemistry and molecular biology

All living organisms use DNA as genetic material, the same genetic code, the same twenty amino acids and ATP as the energy currency. The sequence of amino acids in a protein such as haemoglobin or cytochrome c differs little between closely related species and more between distantly related ones. The proteins of humans and chimpanzees are almost identical, whereas those of humans and fishes differ in many more positions. Blood-protein (serological) tests lead to the same conclusion.

Geographical distribution

The distribution of organisms across the world fits an evolutionary history. Marsupials such as kangaroos are found mainly in Australia, which separated from other continents long ago, while placental mammals occupy the rest of the world. Islands such as Fernando Po (Bioko), near Nigeria, contain species resembling but not identical to those on the mainland, which indicates descent with modification after isolation.

Observed change

Evolution can be seen directly in selective breeding, in the origin of antibiotic-resistant bacteria and in the colour change of peppered moths, all discussed below.

7.3 Theories of evolution

Lamarck's theory

Jean-Baptiste Lamarck, a French biologist, proposed in 1809 the first systematic theory. It rests on two ideas. The first is **use and disuse**: organs that are used frequently become larger and stronger, while those not used degenerate. The second is the **inheritance of acquired characters**: features that an individual acquires during life are passed to its offspring. Lamarck explained the long neck of the giraffe by saying that ancestral giraffes stretched their necks to reach high leaves, that the stretching lengthened the neck slightly, and that the longer neck was inherited by the young.

Experiments do not support the theory. August Weismann cut the tails off mice for twenty-two generations, yet newborn mice always had normal tails. Acquired characters such as scars, muscle development or a tanned skin do not

alter the genes in the gametes. Modern genetics explains why: only changes in the DNA of gametes are inherited.

Darwin's theory of natural selection

Charles Darwin, working in England, and Alfred Russel Wallace independently reached the same theory, which Darwin published in *On the Origin of Species* (1859). It rests on observations and inferences that can be set out in order:

1. **Overproduction.** Organisms produce far more offspring than can survive. A single cod or a single toad lays thousands of eggs, yet the populations remain roughly stable.
2. **Struggle for existence.** Because resources such as food, water, space and mates are limited, individuals compete with one another, with other species and with the environment (predators, disease, climate).
3. **Variation.** Members of a species differ from one another in many characters, and much of this variation is inherited.
4. **Survival of the fittest.** Individuals with variations that suit them better to the environment are more likely to survive and reproduce. “Fittest” means best adapted, not strongest.
5. **Inheritance.** The survivors pass the favourable variations to their offspring. Over many generations the favourable character becomes more common, and the population changes.

Darwin explained the giraffe in the following way. Giraffes in the ancestral population varied in neck length. In times of scarcity, those with longer necks could reach leaves that others could not, so they survived and left more offspring. After many generations the average neck length of the population increased.

Darwin did not know how variation arises or how it is inherited. Mendel's work, and later the discovery of mutation, supplied the missing mechanism. The combination of Darwin's natural selection with genetics is called the **modern synthesis** or neo-Darwinism.

Feature	Lamarck	Darwin
Source of change	Use and disuse of organs	Natural selection acting on existing variation
Acquired characters	Inherited	Not required; only inherited variation matters
Role of environment	Directly induces change in the individual	Selects among variants already present
Modern view	Rejected	Accepted, with genetics added

7.4 Natural selection in action

Variation and its sources

Natural selection can act only if there is heritable variation. Variation arises from mutation (new alleles), from crossing over and independent assortment in meiosis, and from the random fusion of gametes at fertilisation. Mutation is the ultimate source; the other processes shuffle the alleles into new combinations. Variation may be continuous (height, mass, skin colour, controlled by many genes and affected by the environment) or discontinuous (blood group, sex, controlled by one or two genes).

Adaptation

An **adaptation** is a structural, physiological or behavioural feature that helps an organism to survive and reproduce in its particular environment. Examples include the thick waxy cuticle and reduced leaves of cacti and many Sahel plants, which reduce water loss; the streamlined body of a fish; the webbed feet of a duck; the long legs and keen sight of the ostrich; and the camel's ability to tolerate dehydration. **Camouflage** (the chameleon changing colour, a stick insect resembling a twig) and **mimicry** (a harmless species resembling a harmful one) are adaptations against predators.

Industrial melanism

The peppered moth (*Biston betularia*) in England occurs in a pale, speckled form and a dark (melanic) form. Before the Industrial Revolution, trees were covered with pale lichens, and the pale moths were well camouflaged on them, whereas the dark moths were easily seen and eaten by birds. Most moths were pale. As factories burned coal, soot killed the lichens and blackened tree trunks. Now the dark form was camouflaged and the pale form conspicuous, so birds ate more pale moths. The dark allele became common in industrial areas, and in some places over 90% of the moths were dark. When air pollution was reduced in the later twentieth century, pale moths became common again. This is a clear example of natural selection: the environment changed, and the selective advantage changed with it.

Antibiotic and pesticide resistance

When a population of bacteria is exposed to an antibiotic, most individuals die, but a few, which carry a mutation that gives them resistance, survive and multiply. The next generation consists largely of resistant bacteria. The antibiotic did not cause the mutation; it selected the individuals that already

had it. Misuse of antibiotics, for example stopping a course early or taking drugs without a prescription, speeds up this selection and produces dangerous resistant strains, including those causing drug-resistant tuberculosis. In the same way, mosquitoes have become resistant to insecticides such as DDT, and some weeds resist herbicides.

Sickle-cell allele and malaria

In Nigeria and much of tropical Africa, the sickle-cell allele (Hb^S) is relatively common even though two copies cause a severe disease. The explanation is natural selection. Heterozygous people (HbAS) are protected against the worst effects of malaria caused by *Plasmodium falciparum*, so in malarial regions they survive and reproduce better than people with normal haemoglobin (HbAA), who may die of malaria, and better than those with sickle-cell anaemia (HbSS). The allele therefore persists in the population. This is called a **balanced polymorphism** or heterozygote advantage.

Artificial selection

Humans have used selective breeding for thousands of years. Farmers choose the best-yielding cows, the fastest-growing broilers, or maize plants resistant to drought, and breed from them only. Over many generations the chosen character is greatly improved. The various breeds of dog, the N'Dama and White Fulani cattle and the many varieties of cassava are all products of artificial selection. Natural selection and artificial selection work in the same way, but in artificial selection the human breeder, not the environment, decides which individuals reproduce.

WORKED EXAMPLE 7.1

In a population of 200 moths on soot-covered trees, 20 are pale and 180 are dark. Birds eat 60% of the pale moths and 10% of the dark moths before they can breed. Assuming the survivors breed equally, find the percentage of the breeding group that is pale.

Solution: pale eaten = 60% of 20 = 12, so pale survivors = 8. Dark eaten = 10% of 180 = 18, so dark survivors = 162. Total survivors = 8 + 162 = 170. Percentage pale = 8/170 × 100 = 4.7%. Before selection pale moths were 20/200 = 10%, so their proportion has fallen to about half.

7.5 Speciation

A **species** is a group of organisms that can interbreed in nature to produce fertile offspring and that are reproductively isolated from other such groups. A horse and a donkey can mate, but the offspring, a mule, is sterile, so they are separate species. **Speciation** is the formation of new species from an existing one.

The commonest route is **allopatric speciation**, which takes place in stages:

1. A population is divided by a barrier such as a river, mountain range, sea or desert (geographical isolation), so the two groups stop exchanging genes.
2. Mutations occur independently in each group, and different environments favour different characters through natural selection.
3. After many generations the groups become so different genetically that, even if they meet again, they can no longer interbreed or produce fertile offspring. This is **reproductive isolation**, and two new species exist.

Darwin's finches on the Galapagos Islands show the result: one ancestral species from the mainland gave rise to many species with beaks adapted to cracking seeds, catching insects or probing flowers. Speciation can also occur without geographical separation, for example through polyploidy in plants, in which a doubling of chromosome sets makes an individual instantly reproductively isolated from its parent species.

Extinction is the disappearance of all members of a species. It occurs naturally when a species cannot adapt to environmental change, and it is now accelerated by human activities such as habitat destruction, over-hunting and pollution. Endangered Nigerian animals include the Cross River gorilla and the Nigeria-Cameroon chimpanzee.

7.6 The origin of life and the history of organisms

The Earth is about 4.5 billion years old. The earliest life, simple prokaryotic cells resembling bacteria, appeared at least 3.5 billion years ago. The widely discussed **chemical evolution** hypothesis proposes that on the early Earth, with an atmosphere of gases such as methane, ammonia, hydrogen and water vapour, energy from lightning and ultraviolet radiation produced simple organic compounds, which built up in the oceans and eventually organised into the first cells. In 1953 Stanley Miller passed electric sparks through such a mixture of gases and obtained amino acids, which supported the idea that organic molecules can form without life.

The major steps in the history of life can be set out in rough sequence: prokaryotes, then photosynthetic bacteria that released oxygen, then eukaryotic cells, then multicellular organisms; marine invertebrates and fishes; the colonisation of land by plants and then by amphibians; reptiles; the age of dinosaurs; birds and mammals; and finally humans, who appeared only in the

last few hundred thousand years. If the whole history of the Earth were compressed into one year, humans would appear only in the last few minutes of 31 December.

Human evolution

Humans (*Homo sapiens*) share a common ancestor with the African apes, including chimpanzees and gorillas; humans did not descend from living chimpanzees. Fossil evidence, much of it from East and southern Africa, shows a sequence including *Australopithecus* (small-brained, walked upright), *Homo habilis* (made stone tools), *Homo erectus* (larger brain, used fire) and finally *Homo sapiens*. Trends include an increase in brain size from about 450 cm^3 to about 1,350 cm^3, the development of bipedal walking, reduction of jaw and teeth size, and the opposable thumb. These changes are not a straight ladder but a branching tree.

7.7 Further worked examples

WORKED EXAMPLE 7.2

Classify each pair as homologous or analogous: (a) the wing of a bat and the arm of a human; (b) the wing of a butterfly and the wing of a pigeon; (c) the fin of a whale and the forelimb of a dog; (d) the eye of an octopus and the eye of a human.

Solution: (a) homologous, the same bones in the same arrangement modified for different uses; (b) analogous, an insect wing is a fold of the body wall, a bird wing is a modified limb; (c) homologous, both are forelimbs with the pentadactyl plan; (d) analogous, the two eyes are similar in function and design but arose separately in different lineages, an example of convergent evolution.

WORKED EXAMPLE 7.3

A bacterial culture contains 10^{8} cells, of which one in 10^{6} is naturally resistant to an antibiotic. Calculate how many resistant cells there are, and what fraction of the survivors they would form if the antibiotic killed all non-resistant cells.

Solution: resistant cells = $10^{8} \times 10^{-6} = 100$ cells. All the other cells ($10^{8} - 100$) die, so the survivors are the 100 resistant cells, which form 100% of the survivors. When they multiply, the whole new population is resistant, even though they were only 0.0001% of the original culture.

WORKED EXAMPLE 7.4

Explain, using Darwin's ideas, how a population of rats could become resistant to a poison, rat bait, used for many seasons in a Lagos market.

Solution: rats show variation, and a few carry a mutant allele that allows them to break down or tolerate the poison. When the poison is laid, most rats die while the resistant ones survive (survival of the fittest). The survivors breed and pass the allele to their young, so each generation has a higher proportion of resistant rats. After several seasons, the poison no longer controls the population.

Exercise 7.1

1. Define evolution and state why a single individual cannot evolve.
2. Name two homologous structures and state the evidence for evolution that they provide.
3. Distinguish between homologous and analogous structures, with one example each.
4. State two ways in which Darwin's theory differs from Lamarck's.
5. List the main steps in Darwin's theory of natural selection in correct order.
6. Explain how the sickle-cell allele remains common in malarial regions of Nigeria.
7. Describe how industrial melanism in the peppered moth illustrates natural selection.
8. Define a species and explain why a mule does not count as a species formed by a horse and a donkey.

9. Explain the role of geographical isolation in speciation.
10. A population of 500 insects contains 50 insecticide-resistant individuals. After spraying, 90% of the non-resistant insects die and all resistant ones survive. Find the percentage of survivors that are resistant.
11. Why is the study of fossils in rock layers important in studying evolution?
12. Explain why misuse of antibiotics encourages resistant bacteria.

Answers to Exercise 7.1

1. Evolution is the gradual change in inherited characteristics of populations over generations; one individual cannot evolve because its genes are fixed and it is the allele frequency of a population that changes.
2. The forelimb of a human and of a bat (or whale); they show the same bone plan, which indicates common ancestry (comparative anatomy).
3. Homologous: same origin and structure, different function (human arm and bat wing). Analogous: same function, different origin (bird wing and insect wing).
4. Lamarck relied on use and disuse and inheritance of acquired characters; Darwin relied on natural selection of inherited variation, and acquired characters are not inherited.
5. Overproduction, struggle for existence, variation, survival of the fittest, inheritance of favourable variations by offspring.
6. Heterozygous HbAS people resist severe malaria, so they survive and reproduce better in malarial areas, maintaining the allele in the population.
7. Pale moths were camouflaged on clean trees and dark ones on soot-blackened trees; birds ate the conspicuous form, so the dark allele became

common in polluted areas.

8. A species is a group that interbreeds to give fertile offspring; a mule is sterile, so horse and donkey remain separate species.
9. A barrier stops gene flow; each group accumulates different mutations and is selected differently until reproductive isolation occurs.
10. Non-resistant = 450; deaths = 405; survivors non-resistant = 45; total survivors = 45 + 50 = 95; resistant share = 50/95 × 100 = 52.6%.
11. Older fossils lie in deeper layers, so they show the sequence of forms through time and reveal transitional types.
12. Incomplete courses or unnecessary use kill the susceptible bacteria but leave partly resistant ones, which then multiply and spread resistance genes.

Chapter summary

- Evolution is the change in inherited characteristics of populations over generations, leading to new species.
- Evidence comes from fossils, comparative anatomy (homologous, analogous and vestigial structures), embryology, biochemistry, geographical distribution and observed change.
- Lamarck proposed use and disuse and inheritance of acquired characters; this is not supported by evidence.
- Darwin proposed natural selection: overproduction, struggle, variation, survival of the fittest and inheritance of favourable characters.
- Variation arises by mutation, meiosis and random fertilisation; natural selection works on this variation.

- Industrial melanism, antibiotic resistance, insecticide resistance and the sickle-cell allele are examples of natural selection.
- Speciation usually follows geographical isolation, which leads to reproductive isolation.
- Life began as simple prokaryotes more than 3.5 billion years ago; humans share a common ancestor with the apes and evolved in Africa.

Chapter 8: Ecology and the Environment

No organism lives alone. A mango tree in Ibadan depends on soil minerals, rainfall and sunlight; it shelters birds and insects; it is attacked by fungi; and its fallen leaves feed decomposers that return nutrients to the soil. Ecology is the branch of biology that studies these relationships between organisms and their surroundings. It is the science behind farming, fisheries, forestry, conservation and the control of pollution, and it is therefore central to questions facing Nigeria: desertification in the north, erosion in the south-east, oil pollution in the Niger Delta, and the loss of forests across the country.

This chapter covers the components of an ecosystem, the abiotic and biotic factors, feeding relationships and energy flow, nutrient cycles, population ecology, the major habitats and vegetation zones of Nigeria, and the effects of human activity on the environment. The UTME frequently tests food chains, ecological pyramids, symbiotic relationships, nutrient cycles, and pollution, and it often includes calculations of population size and density.

8.1 Basic terms

Ecology is the study of the interrelationships between living organisms and their environment. The **environment** is everything that surrounds and affects an organism, and it has two parts: the **abiotic** (non-living) factors and the **biotic** (living) factors.

- A **habitat** is the place where an organism lives, such as a pond, a termite mound or a rock surface.

- A **population** is all the individuals of one species living in a particular area at a given time.
- A **community** is all the populations of different species living together in a habitat.
- An **ecosystem** is a community together with the non-living environment with which it interacts, as a functioning unit. A pond, a forest and a farm are ecosystems.
- A **biome** is a large region defined by its climate and dominant vegetation, such as the tropical rainforest or the savanna.
- The **biosphere** is the zone of the Earth in which life exists, including the lower atmosphere, the oceans and the upper soil.
- The **niche** of an organism is its role in the ecosystem: what it eats, where it lives, when it is active, and how it interacts with other species.

The levels of ecological organisation therefore run from organism, to population, to community, to ecosystem, to biome, to biosphere.

> **Key point.** Distinguish population (one species) from community (many species). Habitat is the "address" of an organism; niche is its "occupation".

8.2 Abiotic factors

The non-living factors of an environment are grouped as climatic, edaphic (soil) and topographic (relief) factors.

Climatic factors

- **Light** is the source of energy for photosynthesis. Its intensity, duration (photoperiod) and wavelength affect plant growth, flowering and the behaviour of animals, for example nocturnal and diurnal activity.
- **Temperature** affects the rate of enzyme-controlled reactions, so it influences metabolism, growth, distribution and germination. Animals have adaptations such as aestivation (summer dormancy of lungfish and snails in the dry season) and hibernation.
- **Rainfall and humidity** determine the availability of water and hence vegetation type. High humidity reduces transpiration and evaporation.
- **Wind** influences evaporation, pollination, seed dispersal and in extreme cases uproots plants.
- **Atmospheric gases**, especially oxygen and carbon dioxide, are needed for respiration and photosynthesis.

Edaphic factors

The soil is a mixture of mineral particles, humus, water, air and living organisms. Its important features are:

- **Texture**, depending on the proportions of sand, silt and clay. Sandy soil has large particles, drains quickly and holds little water. Clay soil has very fine particles, holds water and mineral ions but drains poorly and is poorly aerated. **Loam**, a mixture of sand, silt and clay with humus, is the best soil for crops because it retains water and nutrients yet drains well.
- **Humus**, the dark decayed organic matter, improves soil structure, holds water and releases minerals.
- **pH**, the acidity or alkalinity of the soil, which affects the availability of nutrients. Most crops prefer a pH of about 6 to 7.

- **Water content** and **mineral salts**, such as nitrates, phosphates and potassium salts.
- **Air spaces**, which supply oxygen to roots and soil organisms.

Topographic factors

Altitude, slope and the direction in which a slope faces affect temperature, rainfall, drainage and erosion. Temperatures fall by about 1°C for every 150 to 200 m of rise, so the Jos Plateau is cooler than the surrounding plains. Water drains down slopes and accumulates in valleys.

8.3 Biotic factors and ecological relationships

Organisms interact with each other in several ways.

Feeding relationships

- **Predation**: a predator (a hawk, lion or lizard) kills and eats another animal, the prey. Predator and prey populations are linked, and a rise in prey numbers is followed by a rise in predators.
- **Competition**: individuals use the same limited resource. *Intraspecific* competition is between members of the same species; *interspecific* competition is between different species, for example weeds and crops competing for light and nutrients.

Symbiosis

Symbiosis means two different species living together in close association. It takes several forms:

- **Mutualism**, where both partners benefit. Examples are nitrogen-fixing *Rhizobium* bacteria in the root nodules of beans and groundnuts, which

receive sugars from the plant while supplying nitrogen compounds; a lichen, in which an alga makes food by photosynthesis while the fungus gives protection and absorbs water; and the cattle egret, which feeds on ticks on the backs of cattle.

- **Commensalism**, where one partner benefits and the other is neither helped nor harmed. Examples are epiphytic orchids growing on trees, and the remora fish attached to sharks.
- **Parasitism**, where one organism (the parasite) benefits and the other (the host) is harmed. Examples include tapeworm, *Plasmodium*, ticks, fleas and the mistletoe plant. An *ectoparasite* lives on the outside of the host; an *endoparasite* lives inside it.

Other feeding relationships include **saprophytism**, in which fungi and bacteria feed on dead organic matter, and **scavenging**, in which animals such as vultures eat carcasses killed by others.

> **Key point.** In mutualism both gain (+ +); in commensalism one gains and one is unaffected (+ 0); in parasitism one gains and one loses (+ −). Associations such as cattle egret and cattle are classed as mutualism.

8.4 Food chains, food webs and ecological pyramids

The sun is the ultimate source of energy for almost all ecosystems. **Producers**, which are green plants and algae, trap light energy by photosynthesis and turn it into chemical energy in carbohydrates. Everything else in the ecosystem depends on this energy.

A **food chain** shows the transfer of energy and materials from one organism to another through feeding. It begins with a producer, followed by consumers: **primary consumers** (herbivores), **secondary consumers** (carnivores that eat herbivores) and **tertiary consumers** (carnivores that eat other carnivores). Each stage is a **trophic level**. An example from a Nigerian savanna is: grass → grasshopper → toad → snake → eagle. In a pond: phytoplankton → zooplankton → tilapia → catfish → fish eagle.

Decomposers, bacteria and fungi, break down dead organisms and wastes, releasing minerals to the soil. They act at every level of the chain. In a real ecosystem most animals eat more than one kind of food, so chains join into a **food web**. A food web is more stable than a food chain because if one species is lost, its consumers can switch to another food.

Energy flow

Energy flows through an ecosystem in one direction, from the sun, through producers to consumers, and finally is lost as heat. At each transfer only about **10%** of the energy passes to the next trophic level. The rest is used by the organism for respiration and movement, lost as heat, or contained in parts that are not eaten (such as bones and roots) or in faeces. For this reason, food chains rarely have more than four or five links, because too little energy remains to support further levels.

Ecological pyramids

An ecological pyramid represents the numbers, biomass or energy at each trophic level, with producers at the base.

- A **pyramid of numbers** shows the number of individuals. It is usually broad at the base and narrow at the top, but can be inverted: a single large tree supporting thousands of insects gives a pyramid whose base is narrow.

- A **pyramid of biomass** shows the dry mass of living material at each level. It is normally upright in land habitats, but may be inverted in an ocean, where the standing mass of fast-reproducing phytoplankton is less than that of the zooplankton that eat it.
- A **pyramid of energy** shows the energy flowing through each level per unit area per year. It is always upright, because energy is lost at each level.

WORKED EXAMPLE 8.1

Producers in a grassland capture 20,000 kJ of energy per square metre per year. Assuming 10% is transferred at each step, calculate the energy available to the secondary consumers, and the percentage of the original energy this represents.

Solution: primary consumers receive 10% of 20,000 = 2,000 kJ. Secondary consumers receive 10% of 2,000 = 200 kJ. As a percentage of the original, 200/20,000 × 100 = 1%.

8.5 Nutrient cycles

Unlike energy, materials are recycled. The main cycles are the water, carbon and nitrogen cycles.

The carbon cycle

The atmosphere contains about 0.04% carbon dioxide. Plants remove CO_2 by photosynthesis and use it to build carbohydrates, fats and proteins. Animals obtain carbon by eating plants or other animals. Carbon returns to the air through respiration of plants, animals and decomposers, through decay, and through the burning (combustion) of wood and fossil fuels such as petroleum,

coal and natural gas. Dead organisms that are buried and compressed over millions of years form fossil fuels, storing carbon. Oceans also absorb and release CO_2. The extra CO_2 released by burning fossil fuels and by clearing forests is a major cause of global warming.

The nitrogen cycle

Nitrogen makes up about 78% of air, but most organisms cannot use it in that form. The cycle makes it usable:

1. **Nitrogen fixation.** Atmospheric nitrogen is converted to ammonium compounds and nitrates by free-living bacteria in the soil (*Azotobacter*), by symbiotic *Rhizobium* in the root nodules of legumes, by cyanobacteria, and by lightning, which oxidises nitrogen in air to oxides that dissolve in rain.
2. **Nitrification.** Ammonium compounds are oxidised by *Nitrosomonas* to nitrites, and nitrites are oxidised by *Nitrobacter* to nitrates, which plants absorb.
3. **Assimilation.** Plants absorb nitrates and build them into amino acids and proteins; animals obtain nitrogen by eating plants.
4. **Ammonification (decay).** Decomposers break down dead matter, urea and faeces, releasing ammonia.
5. **Denitrification.** In waterlogged, oxygen-poor soils, denitrifying bacteria (*Pseudomonas*) convert nitrates back to nitrogen gas, which escapes to the air.

Farmers practise crop rotation, including a legume such as cowpea or groundnut every few seasons, in order to restore nitrogen to the soil.

The water cycle

Water evaporates from oceans, rivers and soil, and is released from plants by transpiration. The vapour rises, cools, condenses into clouds and falls as rain. Water then runs off into rivers, seeps into the ground (infiltration) and is taken up by plants, and the cycle continues.

8.6 Populations

Population density is the number of individuals per unit area or volume. Ecologists seldom count every individual, so they use sampling methods.

- **Quadrat method**, for plants and slow-moving animals. A square frame (often 1 m × 1 m) is placed at random positions, the organisms inside are counted, and the mean count per quadrat is multiplied up to the whole area.
- **Transect**, a line along which organisms are recorded, used to study changes in distribution across a gradient such as from the shore into the sea.
- **Capture–mark–release–recapture**, for mobile animals. A sample is caught, marked harmlessly and released. Later a second sample is caught. The population is estimated by: $N = (M \times C)/R$, where M is the number marked in the first sample, C is the total number caught in the second sample, and R is the number of marked animals recaptured.

WORKED EXAMPLE 8.2

Students in Ilorin place ten quadrats, each 1 m², randomly in a field and count 4, 7, 3, 6, 5, 8, 2, 5, 6 and 4 plants of Siam weed. The field measures 400 m². Estimate the population.

Solution: total counted = 4 + 7 + 3 + 6 + 5 + 8 + 2 + 5 + 6 + 4 = 50. Mean per quadrat = 50/10 = 5 plants per m². Estimated population = 5 × 400 = 2,000 plants.

WORKED EXAMPLE 8.3

In a pond, 60 tilapia are caught, marked and released. A week later 80 fish are caught, of which 12 are marked. Estimate the population of tilapia.

Solution: N = (M × C)/R = (60 × 80)/12 = 4,800/12 = 400 fish.

Factors affecting population size

Population size changes with **natality** (birth rate), **mortality** (death rate), **immigration** (arrival) and **emigration** (departure). Population change = (births + immigrants) − (deaths + emigrants). If these balance, the population is stable.

When a few organisms enter a new habitat with plentiful resources, the population grows slowly at first (lag phase), then rapidly (exponential or log phase), and finally levels off at the **carrying capacity**, the maximum population the environment can support, as food, space or other resources become limiting. This gives an S-shaped (sigmoid) growth curve. Factors that limit population growth include food shortage, predation, disease, competition

and accumulation of wastes. These are *density-dependent* factors because their effect increases as the population becomes denser, while fires, floods and drought act regardless of density.

Succession

Ecological succession is the gradual and orderly replacement of one community by another in an area until a stable climax community is formed. In primary succession, which begins on bare rock or sand, *pioneer* species such as lichens and mosses colonise the surface, then die and form soil. Grasses and herbs follow, then shrubs, and finally trees form the climax. Secondary succession occurs where vegetation has been cleared but soil remains, such as an abandoned farm in a forest zone, and is faster.

8.7 Habitats and vegetation zones of Nigeria

Aquatic habitats

Freshwater habitats (ponds, streams, rivers and lakes) have low salt content, and their organisms, for example tilapia, tadpoles and water lilies, have adaptations for obtaining oxygen dissolved in water. Marine habitats such as the Atlantic coast and the Lagos lagoon have high salinity. Estuarine and mangrove habitats in the Niger Delta have brackish water, with mangrove plants adapted by stilt (prop) roots and breathing roots (pneumatophores).

Terrestrial habitats

Nigeria's vegetation changes from south to north with decreasing rainfall.

Zone	Rainfall and features	Typical vegetation and animals
Mangrove and freshwater swamp	Coastal, waterlogged, very heavy rain	Red mangrove, raffia palms; crabs, mudskippers
Tropical rainforest	Over 1,500 mm; short dry season; tall trees in layers (canopy)	Mahogany, iroko, obeche; monkeys, birds, snakes
Guinea savanna	About 1,000 to 1,500 mm; grass with scattered trees	Locust bean, shea butter tree; antelope, buffalo
Sudan savanna	About 600 to 1,000 mm; shorter grass, fewer trees	Acacia, baobab; hyena, ostrich
Sahel savanna	Under 600 mm; very sparse scrub	Thorny shrubs, desert grasses; camels, lizards

8.8 Human impact: pollution and conservation

Pollution

Pollution is the addition to the environment of substances or energy in amounts that harm living things.

- **Air pollution.** Sources include vehicle exhaust, industrial smoke, gas flaring and the burning of refuse. Carbon monoxide combines with haemoglobin and reduces oxygen transport; sulphur dioxide and nitrogen oxides cause acid rain; CO_2 and methane cause the greenhouse effect and global warming; and CFCs deplete the ozone layer.
- **Water pollution.** Sources include crude-oil spills, untreated sewage, industrial effluent and agricultural runoff of fertiliser. Excess nitrates and phosphates cause **eutrophication**: algae multiply rapidly (an algal bloom), then die; bacteria decomposing them use up the dissolved oxygen, and fish die. Oil spills coat the gills of fish and the feathers of birds and destroy

mangroves in the Niger Delta. Contaminated water spreads cholera and typhoid.

- **Soil pollution.** Heavy metals, pesticides, plastics and oil damage soil organisms and plant growth.
- **Noise pollution.** Excessive noise causes stress and hearing loss.

Erosion, desertification and deforestation

Soil erosion is the removal of topsoil by water or wind. It is encouraged by removal of vegetation, overgrazing, bush burning and farming on slopes, and is severe in parts of Anambra, Imo and Abia states (gully erosion). Control measures include terracing, contour ploughing, planting cover crops, afforestation and building drainage channels. **Desertification** is the spread of desert-like conditions into semi-arid land, driven by overgrazing, felling for firewood and drought, and it is a serious threat in the far north. The planned shelter belt, the Great Green Wall, is meant to check it. **Deforestation**, the clearing of forests for farmland, timber and settlement, causes loss of biodiversity, erosion, flooding and increased atmospheric CO_2.

Conservation

Conservation is the planned management of natural resources to prevent their waste, destruction or loss. Methods include creating national parks and game reserves (Yankari, Gashaka-Gumti, Cross River), afforestation and reforestation, laws that limit hunting and logging, selective felling, recycling, sustainable fishing with legal net sizes, the use of biological control instead of pesticides, and public education.

WORKED EXAMPLE 8.4

A farmer in Kaduna has a field whose soil analysis gives 60% sand, 30% silt and 10% clay by mass. A sample of 250 g is taken. Calculate the mass of each component and say what type of soil this is likely to be.

Solution: sand = 60/100 × 250 = 150 g; silt = 30/100 × 250 = 75 g; clay = 10/100 × 250 = 25 g (check: 150 + 75 + 25 = 250 g). The soil is dominated by sand, so it is a sandy loam: it drains freely, holds relatively little water and needs added humus or irrigation.

Exercise 8.1

1. Distinguish between a habitat and a niche.
2. Define a community and an ecosystem, and give one example of each.
3. Name three edaphic factors and state how each affects plant growth.
4. Classify as mutualism, commensalism or parasitism: (a) tapeworm in the human gut; (b) *Rhizobium* in bean roots; (c) an orchid growing on a tree trunk.
5. Write a food chain of five links from a Nigerian savanna and name the trophic level of each.
6. Explain why food chains rarely have more than five links.
7. A producer level fixes 50,000 kJ m^{-2} $year^{-1}$. How much reaches tertiary consumers if 10% is passed on at each step?
8. Describe the roles of *Nitrosomonas* and *Nitrobacter* in the nitrogen cycle.

9. In a capture–recapture exercise, 45 snails were marked; a second sample of 90 snails included 15 marked ones. Estimate the population.
10. Five quadrats of area 0.5 m² each contain 9, 12, 6, 10 and 8 plants. Estimate the number of plants in an area of 600 m².
11. Explain how excess fertiliser in a river can kill fish.
12. State four methods of controlling soil erosion.

Answers to Exercise 8.1

1. A habitat is the place where an organism lives; a niche is its role or way of life in the ecosystem, including its food, activity and interactions.
2. A community is all the populations living in a habitat (all organisms in a pond); an ecosystem is the community together with its non-living environment (the pond with its water, light and minerals).
3. Texture (controls drainage and water retention), pH (controls nutrient availability), humus (supplies minerals and improves structure); also water content and mineral salts.
4. (a) parasitism; (b) mutualism; (c) commensalism.
5. Grass (producer) → grasshopper (primary consumer) → toad (secondary consumer) → snake (tertiary consumer) → eagle (quaternary consumer).
6. Only about 10% of energy passes to the next level, so very little remains after four or five transfers to support more organisms.
7. Primary consumers: 5,000; secondary: 500; tertiary: 50 kJ m^{-2} $year^{-1}$.
8. *Nitrosomonas* oxidises ammonium compounds to nitrites; *Nitrobacter* oxidises nitrites to nitrates, which plants can absorb.
9. N = (45 × 90)/15 = 4,050/15 = 270 snails.

10. Total = 9 + 12 + 6 + 10 + 8 = 45; area sampled = 5 × 0.5 = 2.5 m²; density = 45/2.5 = 18 per m²; population = 18 × 600 = 10,800 plants.
11. Fertiliser causes an algal bloom; when the algae die, decomposing bacteria use up dissolved oxygen, so fish suffocate (eutrophication).
12. Terracing, contour ploughing, cover crops and afforestation (also drainage channels and reduced overgrazing).

Chapter summary

- Ecology studies the interactions between organisms and their environment; the levels run from organism to population, community, ecosystem, biome and biosphere.
- Abiotic factors are climatic, edaphic and topographic; loam is the best agricultural soil.
- Biotic relationships include predation, competition, mutualism, commensalism and parasitism.
- Energy flows one way through food chains and is lost as heat, with about 10% transferred at each trophic level; pyramids of energy are always upright.
- Carbon, nitrogen and water are recycled; nitrogen cycle stages are fixation, nitrification, assimilation, ammonification and denitrification.
- Population size is estimated with quadrats and capture–recapture, N = (M × C)/R; growth levels off at the carrying capacity.
- Nigeria's vegetation grades from mangrove and rainforest in the south to savanna zones and Sahel in the north.

- Pollution, erosion, desertification and deforestation damage the environment, and conservation seeks to manage resources sustainably.

Chapter 9: Plant Biology, Growth and Coordination

Plants support almost all life on land. They produce the oxygen we breathe and the carbohydrates we eat, from the yam and cassava on our farms to the cocoa, oil palm and rubber that have shaped the Nigerian economy. Earlier chapters have introduced photosynthesis, osmosis and transport in plants. This chapter brings these ideas together by looking at the structure of the flowering plant, the details of photosynthesis and the factors that control it, mineral nutrition, growth and germination, and the way plants respond to their environment through hormones. The chapter then turns briefly to coordination in animals, the nervous system and the sense organs, so that the idea of "response to stimuli" is seen in both kingdoms.

9.1 Structure of the flowering plant

A flowering plant (angiosperm) has a **root system** and a **shoot system**. The shoot consists of the stem, leaves, buds, flowers and fruits. Flowering plants are divided into two classes whose differences are examined frequently.

Feature	Monocotyledons	Dicotyledons
Seed leaves (cotyledons)	One	Two
Leaf venation	Parallel	Net-like (reticulate)
Root system	Fibrous	Tap root with branches
Vascular bundles in stem	Scattered	Arranged in a ring
Cambium	Absent, so no secondary thickening	Present, so stems thicken
Flower parts	In threes or multiples of three	In fours or fives or their multiples
Examples	Maize, rice, guinea corn, palm	Beans, hibiscus, mango, orange

Plant tissues

Plant tissues are of two kinds. **Meristematic tissue** consists of small cells with dense cytoplasm and thin walls that divide continually. It is found at the tips of roots and shoots (apical meristems), which cause growth in length, and in the cambium, which causes growth in girth. **Permanent tissues** are made of cells that have stopped dividing. They include:

- **Epidermis**, the outer protective layer, often covered by a waxy cuticle on leaves and stems;
- **Parenchyma**, thin-walled living cells that store food and fill spaces; the photosynthetic cells of the leaf are specialised parenchyma;
- **Collenchyma**, with thickened corners, giving flexible support to young stems;
- **Sclerenchyma**, dead cells with walls thickened by lignin, giving strong support (fibres and stone cells);

- **Xylem**, which conducts water and mineral salts upward and supports the plant; its vessels are dead, hollow, lignified tubes;
- **Phloem**, which conducts manufactured food (sucrose and amino acids) in both directions; it consists of sieve tubes and companion cells.

The root

A young root has, from outside to inside, the piliferous layer bearing root hairs, the cortex, the endodermis, the pericycle and the central vascular tissue in which xylem and phloem lie on different radii. Root hairs are extensions of epidermal cells that greatly increase the surface area for absorbing water and mineral ions. The root fixes the plant in the soil, absorbs water and minerals, and in some plants stores food (carrot, cassava) or acts as a prop (maize and mangrove).

The stem and leaf

In a dicot stem the vascular bundles are in a ring, with phloem outside and xylem inside and a layer of cambium between them. The stem supports leaves and flowers, carries materials between root and leaf and sometimes stores food (the sugarcane, the yam tuber) or reproduces vegetatively.

The leaf is the main organ of photosynthesis. A section through a dicot leaf shows the following layers: a **waxy cuticle** and **upper epidermis** (transparent, to allow light through); the **palisade mesophyll**, columnar cells packed with chloroplasts and arranged upright to catch light; the **spongy mesophyll**, with irregular cells and large air spaces for gas circulation; the **vascular bundles** (veins) bringing water and removing sugars; and the **lower epidermis**, with many **stomata** each bounded by two guard cells. The leaf is thin and flat, giving a large surface area and a short diffusion path.

9.2 Photosynthesis in detail

Chapter 3 introduced photosynthesis as the process by which green plants make carbohydrates from carbon dioxide and water using light. The overall equation is:

$6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2$ (in the presence of light and chlorophyll)

The two stages

- The **light-dependent stage** takes place in the grana of the chloroplast. Chlorophyll absorbs light, mainly red and blue wavelengths, and reflects green, which is why leaves look green. Light energy is used to split water molecules (**photolysis**): water gives hydrogen, which is carried by a carrier molecule, and oxygen, which is released as a by-product. ATP is also formed.
- The **light-independent stage** (dark stage) takes place in the stroma. The hydrogen and ATP from the first stage are used to reduce carbon dioxide to carbohydrate. Light is not directly needed here, but the stage depends on the products of the first.

The glucose made is used for respiration, converted to sucrose for transport, converted to starch for storage, or used to build cellulose, and with nitrates to make amino acids and proteins.

Factors affecting the rate of photosynthesis

The rate of photosynthesis depends on light intensity, carbon dioxide concentration, temperature and the availability of water and chlorophyll. When one of these is in short supply it limits the rate, and it is called a

limiting factor. Blackman's principle states that when a process depends on several factors, its rate is limited by the factor that is nearest its minimum value.

- **Light intensity.** The rate rises in proportion to light intensity until some other factor becomes limiting, after which the curve levels off.
- **Carbon dioxide.** Air contains only 0.04% CO_2, so it is often limiting. Raising the concentration, as in greenhouses, increases the rate until another factor limits.
- **Temperature.** Because the dark stage is controlled by enzymes, the rate increases with temperature up to an optimum of about 25 to 35°C, then falls sharply as enzymes are denatured above about 40°C.

Experiments on photosynthesis

- **Test for starch in a leaf.** The leaf is dipped in boiling water to kill the cells and stop reactions, then boiled in ethanol in a water bath to remove chlorophyll, rinsed in warm water to soften it, and covered with iodine solution. Blue-black shows starch is present.
- **Light is necessary.** A leaf partly covered with black paper or foil gives starch only in the uncovered part.
- **Chlorophyll is necessary.** A variegated leaf (such as croton) gives starch only in the green areas.
- **Carbon dioxide is necessary.** A plant placed with sodium hydroxide or potassium hydroxide solution (which absorbs CO_2) in a closed container does not make starch, but a control with water does.
- **Oxygen is released.** A water plant such as *Elodea* under an inverted funnel gives bubbles of gas that relight a glowing splint.

Before such an experiment the plant is **destarched** by keeping it in darkness for 24 to 48 hours so that any starch found afterwards must have been made during the experiment.

Key point. In a photosynthesis experiment, the "control" has every condition except the factor under test. Always destarch before testing for starch, and decolourise with hot ethanol before adding iodine.

WORKED EXAMPLE 9.1

The equation for photosynthesis is $6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2$. How many grams of glucose can be formed from 264 g of carbon dioxide? (C = 12, H = 1, O = 16)

Solution: molar mass of CO_2 = 12 + 32 = 44 g/mol. Moles of CO_2 = 264/44 = 6 mol. Six moles of CO_2 give one mole of glucose. Molar mass of glucose = (6 × 12) + (12 × 1) + (6 × 16) = 72 + 12 + 96 = 180 g/mol. Mass of glucose = 1 × 180 = 180 g. (Also 6 mol of oxygen, 192 g, is released.)

9.3 Mineral nutrition and soil fertility

Plants absorb mineral ions from the soil by active transport. Elements needed in large amounts are called **macronutrients**: nitrogen, phosphorus, potassium, calcium, magnesium and sulphur. Those needed in tiny amounts are **micronutrients** or trace elements, such as iron, manganese, zinc, copper and boron.

Element	Use in the plant	Deficiency symptom
Nitrogen	Amino acids, proteins, chlorophyll	Stunted growth; yellowing of older leaves (chlorosis)
Phosphorus	ATP, nucleic acids, cell membranes; root growth	Poor root growth; purple colouring of leaves
Potassium	Enzyme activation, opening of stomata	Yellow leaf edges with dead patches
Magnesium	Part of the chlorophyll molecule	Yellowing between veins of older leaves
Iron	Formation of chlorophyll	Yellowing of young leaves
Calcium	Cell wall (calcium pectate), cell division	Poor growth of root and shoot tips

Soil fertility is maintained by adding **manure** (organic, improves structure and releases nutrients slowly) or **inorganic fertilisers** such as NPK, practising crop rotation with legumes, fallowing, and mulching. Hydroponics, growing plants in a nutrient solution without soil, is used to study mineral deficiency. Parasitic and insectivorous plants meet some needs differently: the sundew traps insects to obtain nitrogen, and *Striga* (witchweed), which attacks sorghum and maize, takes its food from the host.

9.4 Growth, seeds and germination

Growth is a permanent increase in size, dry mass or cell number. It involves cell division, cell enlargement and differentiation. Plants grow throughout life at their meristems, while most animals grow only until maturity. Growth is measured by length, fresh mass or, most reliably, **dry mass**, since fresh mass changes with water content. A graph of growth against time is S-shaped.

A seed consists of a protective **testa**, an **embryo** (plumule, radicle, and the cotyledons) and food stores in the cotyledons or the endosperm. **Germination** is the resumption of growth by the embryo. A viable seed needs three conditions: **water** (to activate enzymes and soften the testa), **oxygen** (for respiration) and a suitable **temperature**. Light is not needed for most seeds. During germination enzymes such as amylase digest stored starch to sugar, which supplies energy and materials to the embryo; because the seedling has no leaves, its dry mass at first falls even though its fresh mass rises. The radicle emerges first and becomes the root, and the plumule grows into the shoot. In **epigeal germination** (bean, cotton, groundnut) the cotyledons are carried above the soil by the elongating hypocotyl. In **hypogeal germination** (maize, cowpea, oil palm) the cotyledons stay beneath the soil because the epicotyl lengthens.

Some seeds are **dormant**: they will not germinate even in favourable conditions until a resting period, a hard seed coat or chemical inhibitors are overcome.

WORKED EXAMPLE 9.2

Ten seedlings of maize have a total dry mass of 0.90 g on day 0 and 2.70 g on day 14. Calculate the percentage increase in dry mass and the average daily increase per seedling.

Solution: increase = 2.70 − 0.90 = 1.80 g. Percentage increase = 1.80/0.90 × 100 = 200%. Average daily increase for all ten = 1.80/14 = 0.1286 g/day, so per seedling = 0.1286/10 = 0.0129 g/day (about 0.013 g per seedling per day).

9.5 Plant hormones and responses

Plants respond to stimuli by growth movements, not by locomotion. A **tropism** is a growth response of part of a plant in a direction determined by the direction of the stimulus. Growth towards the stimulus is positive; away from it is negative.

- **Phototropism**, response to light: stems are positively phototropic, roots are usually negatively phototropic.
- **Geotropism (gravitropism)**, response to gravity: roots are positively geotropic, shoots are negatively geotropic.
- **Hydrotropism**, response to water: roots grow towards water.
- **Thigmotropism**, response to touch: tendrils of climbing plants such as the yam vine coil round supports.
- **Chemotropism**, response to chemicals: the growth of the pollen tube towards the ovule.

A **nastic movement** is a response not determined by the direction of the stimulus, such as the folding of the leaflets of *Mimosa pudica* (the sensitive plant) when touched, or the opening and closing of flowers.

Tropisms are controlled by growth substances (hormones). The most important is **auxin** (indole acetic acid, IAA), made in the tips of shoots and roots. Auxin moves down the stem and stimulates cell elongation. In a shoot lit from one side, auxin moves to the shaded side, where cells elongate more, so the shoot bends towards the light. In roots, high auxin concentration inhibits elongation, so a root lying horizontally has more auxin on its lower side, which slows growth there and the root bends downward. Experiments

with oat or maize coleoptiles show that removing the tip stops growth and bending, and that placing a tip on one side of a decapitated coleoptile makes it curve.

Other plant hormones are **gibberellins** (stem elongation, breaking of seed dormancy), **cytokinins** (cell division, delay of leaf senescence), **abscisic acid** (closing of stomata, dormancy) and **ethene (ethylene)**, which ripens fruits and causes leaf fall. Synthetic auxins such as 2,4-D are used as selective weedkillers because they cause broad-leaved weeds to overgrow and die, and rooting powders for cuttings contain auxin.

9.6 Coordination in animals

Animals respond to stimuli much faster than plants because they have a nervous system and endocrine (hormonal) glands.

The nervous system

The **central nervous system** (CNS) comprises the brain and the spinal cord. The **peripheral nervous system** consists of the nerves that connect the CNS to receptors and effectors. The functional unit is the **neurone**, which has a cell body with a nucleus, branching dendrites that receive impulses, and a long axon that carries impulses away, often wrapped in a fatty myelin sheath that speeds conduction. Three kinds of neurone exist: **sensory neurones** carry impulses from receptors to the CNS, **relay (intermediate) neurones** connect neurones within the CNS, and **motor neurones** carry impulses from the CNS to effectors (muscles and glands). Neurones meet at a **synapse**, a tiny gap across which a chemical transmitter (acetylcholine in many cases) diffuses to start an impulse in the next neurone. Synapses ensure that impulses travel in only one direction.

The brain has three main parts: the **cerebrum** (thought, memory, voluntary movement and the interpretation of senses), the **cerebellum** (balance and coordination of movement) and the **medulla oblongata** (controls involuntary actions such as heartbeat, breathing and swallowing). The **hypothalamus** regulates temperature and links the nervous and endocrine systems.

A **reflex action** is a rapid, automatic response to a stimulus that does not involve conscious thought. The path is the **reflex arc**: stimulus → receptor → sensory neurone → relay neurone in the spinal cord → motor neurone → effector → response. Removing the hand from a hot pot is an example. Voluntary actions, such as writing, involve the cerebrum.

The endocrine system

Hormones are chemical messengers produced by ductless glands, carried in the blood, and acting on target organs. They are slower but longer-lasting in effect than nervous impulses. Examples are insulin and glucagon (pancreas, blood sugar), thyroxine (thyroid, metabolic rate), adrenaline (adrenal glands, the “fight or flight” response), and the sex hormones described in Chapter 6. Failure of the endocrine glands causes disorders such as diabetes mellitus (too little insulin) and goitre (lack of iodine, which prevents thyroxine formation).

The sense organs

The **eye** is the organ of sight. Light passes through the **cornea**, aqueous humour, the **pupil** (opening in the iris), the **lens** and the vitreous humour to form an inverted image on the **retina**, which contains rods (dim light, black-and-white vision) and cones (bright light, colour vision). The **optic nerve** carries impulses to the brain. The iris controls pupil size: in bright light the circular muscles contract, narrowing the pupil, and in dim light the radial muscles contract, widening it. **Accommodation** is the adjustment of the lens:

for a near object, the ciliary muscles contract, the suspensory ligaments slacken and the lens becomes thicker; for a distant object the reverse occurs. Short sight (myopia) is corrected with a concave lens, long sight (hypermetropia) with a convex lens.

The **ear** is the organ of hearing and balance. Sound waves are collected by the pinna and travel down the ear canal to vibrate the eardrum. Three small bones (malleus, incus and stapes) amplify the vibrations to the oval window of the cochlea in the inner ear, where hair cells convert the movement of fluid into impulses carried by the auditory nerve. The semicircular canals detect changes in the position and movement of the head, and the Eustachian tube equalises pressure on the two sides of the eardrum. The **skin** contains receptors for touch, pressure, pain, heat and cold; the **tongue** has taste buds (sweet, sour, salty, bitter, and savoury); and the **nose** detects airborne chemicals through the olfactory epithelium.

WORKED EXAMPLE 9.3

Order the following structures to show the path of a reflex action when a person touches a hot object: motor neurone, effector muscle, receptor in skin, sensory neurone, relay neurone, spinal cord.

Solution: receptor in skin → sensory neurone → relay neurone within the spinal cord → motor neurone → effector muscle. The brain is not necessary for the reflex, though the person later becomes aware of the pain through impulses that pass to the brain.

WORKED EXAMPLE 9.4

A student places a potted seedling by a window. After three days, the shoot is bent towards the window. Explain the observation.

Solution: the shoot tip makes auxin, which moves down to the zone of elongation. Light from the window causes auxin to concentrate on the side away from the light. Cells on the shaded side elongate more than those on the lit side, so the shoot curves towards the light. The response is positive phototropism, and it helps the leaves to receive more light for photosynthesis.

Exercise 9.1

1. State four differences between monocotyledons and dicotyledons.
2. Name the tissue that transports water in plants and the tissue that transports sucrose.
3. Explain why palisade cells are suited to photosynthesis.
4. Write the overall equation for photosynthesis and name the two stages.
5. What is photolysis and where does it occur?
6. Calculate the mass of glucose produced from 108 g of water in photosynthesis. (H = 1, C = 12, O = 16)
7. Why is a plant destarched before an experiment? How is this done?
8. Name two mineral elements needed for the formation of chlorophyll, and state the symptom of deficiency of nitrogen.
9. List the conditions necessary for germination, and distinguish epigeal from hypogeal germination.

10. Explain how auxin causes a shoot to bend towards light.
11. Distinguish between a tropism and a nastic movement.
12. Describe the path of a reflex arc, and state one difference between a reflex and a voluntary action.

Answers to Exercise 9.1

1. Any four of: one versus two cotyledons; parallel versus net venation; fibrous versus tap roots; scattered versus ringed vascular bundles; no cambium versus cambium; floral parts in threes versus fours or fives.
2. Xylem transports water; phloem transports sucrose.
3. They are tall, closely packed cells near the upper surface, with many chloroplasts, so they absorb most of the light.
4. $6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2$; the light-dependent and light-independent stages.
5. Photolysis is the splitting of water by light energy to give hydrogen and oxygen; it occurs in the grana of the chloroplasts.
6. Molar mass H_2O = 18 g/mol; 108/18 = 6 mol; 6 mol of water give 1 mol of glucose = 180 g.
7. So that any starch found was made during the experiment; the plant is kept in the dark for 24 to 48 hours.
8. Magnesium and iron (nitrogen also forms part of chlorophyll); nitrogen deficiency gives stunted growth and yellowing of leaves.
9. Water, oxygen and suitable temperature; in epigeal germination the cotyledons rise above the soil, in hypogeal they stay below it.
10. Auxin concentrates on the shaded side, causing greater cell elongation there, so the shoot bends towards the light.

11. A tropism is a growth movement whose direction depends on the direction of the stimulus; a nastic movement does not depend on direction (for example the closing of Mimosa leaves).
12. Receptor → sensory neurone → relay neurone → motor neurone → effector; a reflex is rapid and automatic without conscious thought, while a voluntary action is controlled by the cerebrum.

Chapter summary

- Flowering plants are monocotyledons or dicotyledons; their tissues include meristems, epidermis, parenchyma, xylem and phloem.
- The leaf is built for photosynthesis, with a thin lamina, palisade and spongy mesophyll and stomata.
- Photosynthesis has a light-dependent stage (photolysis, ATP) and a light-independent stage (reduction of CO_2); its rate is limited by light, CO_2 and temperature.
- Plants need macronutrients and micronutrients; deficiency causes characteristic symptoms such as chlorosis.
- Germination needs water, oxygen and warmth; it may be epigeal or hypogeal.
- Auxin controls tropisms by stimulating cell elongation; other hormones include gibberellins, cytokinins, abscisic acid and ethene.
- Animals respond through the nervous system (neurones, reflex arc, brain) and the endocrine system (hormones).
- The eye forms images on the retina and the ear detects sound and balance.

Chapter 10: Microorganisms, Disease and Public Health

Most of the living things on Earth cannot be seen without a microscope. Bacteria, many fungi, protozoa, algae and viruses make up the world of **microorganisms**, and their effect on human affairs is enormous. They turn milk into yoghurt and cassava into garri, they break down dead plants and animals, and they provide antibiotics. They also cause some of the diseases that kill most Nigerians: malaria, cholera, typhoid, tuberculosis and HIV/AIDS. This final chapter describes the major groups of microorganisms, their beneficial and harmful activities, the way in which diseases spread and are prevented, the body's defences, and the basic principles of public health and biotechnology.

The UTME regularly asks for the causative agent, vector, symptoms and control of named diseases, for the uses of microorganisms in industry, and for the meaning of terms such as pathogen, vector, immunity, vaccine, antibiotic and epidemic. A clear table of each disease is the best revision tool, and several are given below.

10.1 The groups of microorganisms

Bacteria

Bacteria are unicellular prokaryotes with no nucleus, a cell wall of peptidoglycan, and a single circular chromosome, and many also carry small rings of DNA called plasmids. They reproduce asexually by **binary fission**, and under good conditions a bacterium can divide every 20 minutes. They are

classified by shape: **cocci** (spherical, such as *Streptococcus*), **bacilli** (rod-shaped, such as *Salmonella*), **spirilla** (spiral) and **vibrios** (comma-shaped, such as *Vibrio cholerae*). Some form resistant **spores** that survive heat and drying. Most bacteria are heterotrophs, either saprophytes or parasites, but some, like cyanobacteria, photosynthesise, and others (nitrifying bacteria) obtain energy from inorganic compounds.

Fungi

Fungi are eukaryotes with cell walls of chitin that have no chlorophyll and feed saprophytically or parasitically. Yeast is unicellular and reproduces by budding. Moulds (*Mucor*, *Rhizopus*, *Penicillium*) and mushrooms are made of thread-like **hyphae** forming a **mycelium**, and they reproduce by spores. Fungi secrete digestive enzymes onto their food and absorb the products.

Protozoa

Protozoa are unicellular eukaryotic organisms. *Amoeba* moves by pseudopodia, *Paramecium* by cilia and *Euglena* by flagellum. Parasitic protozoa include *Plasmodium* (malaria), *Trypanosoma* (sleeping sickness) and *Entamoeba histolytica* (amoebic dysentery).

Viruses

Viruses are not cells. Each consists of genetic material (DNA or RNA) inside a protein coat (the capsid), sometimes inside an outer envelope. They are far smaller than bacteria, they show no signs of life outside a host, and they reproduce only inside living cells, by taking over the cell's machinery. Because they are not affected by antibiotics, they are controlled by vaccines and antiviral drugs.

Group	Cell type	Typical size	Disease examples
Bacteria	Prokaryote	0.5 to 5 µm	Cholera, typhoid, tuberculosis, tetanus
Fungi	Eukaryote	Variable	Ringworm, athlete's foot, candidiasis
Protozoa	Eukaryote	10 to 100 µm	Malaria, sleeping sickness, amoebic dysentery
Viruses	Acellular	20 to 300 nm	Polio, measles, HIV/AIDS, Lassa fever, influenza

10.2 Useful activities of microorganisms

- **Decomposition and recycling.** Saprophytic bacteria and fungi break down dead matter, releasing carbon dioxide and minerals (Chapter 8). Without them waste would accumulate and nutrients would be locked away.
- **Nitrogen fixation and nitrification** by soil bacteria maintain soil fertility.
- **Food production.** Yeast is used in baking and brewing because it ferments sugar to ethanol and carbon dioxide ($C_6H_{12}O_6 \rightarrow 2C_2H_5OH + 2CO_2$); the gas makes dough rise. Lactic acid bacteria turn milk into yoghurt and cheese and ferment cassava for garri and ogi. Fermented locust beans (iru, dawadawa) and palm wine involve bacteria and wild yeasts. Mushrooms are eaten directly.
- **Medicine.** Antibiotics are chemicals made by microorganisms that kill or stop the growth of bacteria. Alexander Fleming discovered penicillin in 1928 when he noticed that the mould *Penicillium* prevented bacterial growth on a culture plate. Streptomycin is made by the bacterium *Streptomyces*. Vaccines and some hormones are also produced with microbial help.
- **Industry and biotechnology.** Microbes are used to make vinegar (*Acetobacter*), citric acid (*Aspergillus*), enzymes for washing powders,

biogas (methane produced by bacteria from animal and plant waste), and in sewage treatment. In genetic engineering, bacteria carrying human genes are used to produce insulin and growth hormone.

- **Digestion.** Bacteria in the gut of ruminants digest cellulose, and bacteria in the human colon make vitamin K.

10.3 Harmful activities of microorganisms

- They cause diseases in humans, animals and crops. Examples in plants include cassava mosaic virus, cocoa swollen shoot virus, and rust and smut fungi in cereals.
- They **spoil food**. Moulds, yeasts and bacteria break down stored food, and some produce toxins; *Aspergillus flavus* makes aflatoxin on groundnuts and maize stored in damp conditions, which can cause liver damage and cancer. *Clostridium botulinum* in poorly canned food causes botulism.
- They cause denitrification in waterlogged soils, reducing fertility.
- They cause decay of wood, leather and textiles.

Food preservation

Food is preserved by preventing the growth of microorganisms. Methods include drying (smoked or sun-dried fish, dried pepper, beans), salting and sugaring (which draw water out of the microbes by osmosis), smoking, refrigeration and freezing (low temperatures slow growth), canning (heating in sealed containers), pasteurisation (milk heated to about 72°C for 15 seconds, then rapidly cooled, which kills disease-causing organisms without spoiling taste), pickling in vinegar, and chemical preservatives such as sodium benzoate.

10.4 Diseases and their transmission

A **disease** is a condition that disturbs the normal functioning of the body. A **pathogen** is an organism that causes disease. Diseases are **infectious** (communicable) when caused by pathogens and passed from one person to another, or **non-infectious** (such as sickle-cell anaemia, kwashiorkor, diabetes, cancer and scurvy), which are inherited, nutritional or caused by organ failure. An **epidemic** is a sudden outbreak affecting many people in a region; a **pandemic** is an epidemic that spreads across many countries; an **endemic** disease is always present in a region, as malaria is in Nigeria. The **incubation period** is the time between infection and the first symptoms. A **carrier** harbours a pathogen without showing symptoms but can pass it on, and a **vector** is an organism that carries a pathogen from one host to another.

Methods of transmission

- **Direct contact**, including touching, kissing and sexual contact (ringworm, scabies, gonorrhoea, HIV).
- **Droplets and air**, by coughing and sneezing (tuberculosis, influenza, measles, meningitis, COVID-19).
- **Contaminated food and water** (cholera, typhoid, dysentery, polio, hepatitis A, and parasitic worms).
- **Vectors**, such as the female *Anopheles* mosquito (malaria), the tsetse fly (sleeping sickness), the housefly (mechanical carrier of cholera and typhoid), the *Simulium* blackfly (river blindness) and fleas (plague).
- **Body fluids**, through blood transfusion and unsterilised needles (HIV, hepatitis B), or from mother to child through the placenta and breast milk.
- **Animal bites and contact**, such as rabies from the bite of an infected dog, and Lassa fever from food or objects contaminated by the urine and

droppings of the multimammate rat.

10.5 Some important diseases

Disease	Causative agent	Transmission	Key symptoms	Control
Malaria	*Plasmodium* (protozoan)	Bite of female *Anopheles* mosquito	Recurring fever, chills, sweating, anaemia, headache	Nets, insecticides, drain stagnant water, prophylactic drugs, ACT treatment
Cholera	*Vibrio cholerae* (bacterium)	Contaminated water and food	Severe watery diarrhoea, vomiting, dehydration	Clean water, sanitation, oral rehydration therapy, vaccine
Typhoid	*Salmonella typhi* (bacterium)	Contaminated food and water; flies	Prolonged fever, abdominal pain, rash	Hygiene, vaccination, antibiotics
Tuberculosis	*Mycobacterium tuberculosis*	Droplets	Persistent cough, blood in sputum, weight loss	BCG vaccine, drugs, ventilation
Poliomyelitis	Poliovirus	Faecal–oral	Paralysis of limbs	Oral or injected polio vaccine
Measles	Measles virus	Droplets	Fever, rash, red eyes	Vaccination
HIV/AIDS	Human immunodeficiency virus	Body fluids, sex, mother to child	Failing immunity, opportunistic infections	Safe sex, screened blood, sterile needles,

Disease	Causative agent	Transmission	Key symptoms	Control
				antiretroviral drugs
Lassa fever	Lassa virus	Rodents' excreta, contact with patients	Fever, bleeding, weakness	Rodent control, storage of food, hygiene, early treatment
Sleeping sickness	*Trypanosoma* (protozoan)	Tsetse fly	Fever, drowsiness, coma	Fly control, clearing bush
Tetanus	*Clostridium tetani*	Deep wounds contaminated with soil	Muscle stiffness, lockjaw	Vaccination, wound care
Ringworm	Fungi (*Trichophyton*)	Contact	Itchy circular patches on skin	Hygiene, antifungal cream

The malaria parasite

Malaria is so common that its life cycle deserves detail. When an infected female *Anopheles* mosquito bites a person, it injects *Plasmodium* sporozoites with its saliva. These travel to the liver, where they multiply, and then invade red blood cells. In the red cells they divide and then burst the cells, releasing a new generation of parasites together with toxins; this happens at regular intervals and causes the recurring bouts of fever. Some parasites develop into gametocytes, which are taken up when another mosquito bites the patient. In the mosquito's gut, they undergo sexual reproduction, and the new sporozoites move into the salivary glands, ready to infect the next person. The mosquito is therefore the **vector** and the final (definitive) host, while the human is the

intermediate host. The mosquito breeds in stagnant water, so control includes draining or oiling pools, clearing gutters, using insecticide-treated nets and sleeping under them, and covering water containers.

Parasitic worms

- **Tapeworm** (*Taenia*) is acquired by eating undercooked beef or pork containing larval cysts; it attaches to the gut with hooks and suckers and absorbs digested food.
- **Roundworm** (*Ascaris*) spreads through eggs in contaminated food or soil.
- **Hookworm** larvae penetrate bare feet from contaminated soil and cause anaemia; wearing shoes prevents this.
- **Schistosoma** (bilharzia) uses freshwater snails as intermediate hosts, and its larvae penetrate the skin of people who bathe or wash in infested water.
- **Guinea worm** (*Dracunculus*) is acquired by drinking water containing infected water fleas; Nigeria has been certified free of it, because of water filtration and treatment of ponds.

10.6 The body's defences and immunity

The body has several lines of defence against pathogens.

- **Barriers:** the skin, mucus in the nose and respiratory tract, which traps microbes, with cilia that sweep them away; hydrochloric acid in the stomach; lysozyme in tears and saliva; and the natural flora of harmless bacteria on skin and gut.
- **Blood clotting** seals wounds and prevents entry of microbes.
- **White blood cells. Phagocytes** engulf and digest pathogens (phagocytosis). **Lymphocytes** produce **antibodies**, proteins that are

specific to a particular **antigen** (a foreign molecule on a pathogen). Antibodies clump pathogens together, neutralise toxins and mark microbes for destruction. After an infection, some lymphocytes remain as **memory cells**, so a second infection by the same pathogen is met by a faster and larger response.

Immunity is the ability of the body to resist a particular disease. It is classified as follows.

Type	How obtained	Duration
Natural active	Recovery from the disease	Long-lasting
Artificial active	Vaccination: dead, weakened or part of a pathogen is injected	Long-lasting
Natural passive	Antibodies from mother via placenta and breast milk	Short (months)
Artificial passive	Injection of ready-made antibodies (antiserum), such as for tetanus or snake bite	Short

In **active** immunity the person makes his or her own antibodies; in **passive** immunity the antibodies are supplied from outside, so memory cells are not formed. A **vaccine** contains an antigen but does not cause the disease. It triggers the production of antibodies and memory cells. When enough of a population is vaccinated, even the unvaccinated gain protection, as the pathogen cannot spread easily (herd immunity). Nigeria's immunisation schedule gives BCG, polio, pentavalent (diphtheria, pertussis, tetanus, hepatitis B and Hib), measles and other vaccines to infants.

Antibiotics kill bacteria but not viruses. **HIV** attacks and destroys helper T-lymphocytes, which coordinate the immune response, so a patient develops AIDS (acquired immune deficiency syndrome) and becomes prone to

infections and some cancers.

> **Key point.** Vaccines prevent disease by giving active immunity. Antibiotics treat bacterial disease but have no effect on viruses. Misusing antibiotics causes resistance (Chapter 7).

10.7 Public health and hygiene

Public health measures prevent disease in whole communities.

- **Provision of safe water**: boiling, filtering, chlorination, and protection of wells and boreholes from contamination by latrines.
- **Proper sewage and refuse disposal**: sealed latrines, septic tanks and treatment works, regular waste collection and burying or incineration of refuse.
- **Vector control**: draining swamps, spraying insecticides, using nets, covering foods against flies.
- **Personal hygiene**: regular bathing, washing hands with soap after using the toilet and before eating, brushing teeth, wearing shoes.
- **Food hygiene**: washing vegetables, cooking meat thoroughly, covering food, storing it in cool conditions.
- **Immunisation, health education, quarantine** (isolating those who may be infected) and **disease surveillance**.
- **Sanitary inspection** of markets, abattoirs and eating places.

Methods of sterilisation

Sterilisation destroys all microorganisms, while **disinfection** reduces their number to a safe level. Heat is the commonest agent: boiling for 10 minutes kills most organisms but not all spores, and an autoclave (steam under pressure at 121°C for 15 minutes) kills everything. Chemical disinfectants include chlorine and alcohol, and ultraviolet light is used on surfaces.

10.8 Culturing and counting microorganisms

Bacteria and fungi are grown in the laboratory on **culture media** such as nutrient agar, a jelly containing nutrients in a Petri dish. **Aseptic technique** is used: the dish and tools are sterilised, the inoculating loop is flamed, the dish is opened as little as possible and is sealed with tape, and incubated upside down at about 25°C so that harmful pathogens are not encouraged to grow at body temperature. Each bacterium that lands on the medium grows into a visible **colony**. Counting colonies allows the number of living bacteria in a sample to be estimated.

Bacterial growth is described by the doubling time. If the number of bacteria doubles every generation time g, then after time t the number is $N = N_0 \times 2^n$, where $n = t/g$.

WORKED EXAMPLE 10.1

A single bacterium divides every 20 minutes in a nutrient broth. How many bacteria will be present after 3 hours, assuming no deaths?

Solution: 3 hours = 180 minutes, so $n = 180/20 = 9$ generations. $N = 1 \times 2^9 = 512$ bacteria.

WORKED EXAMPLE 10.2

1 cm^3 of a milk sample is diluted 1000 times, and 0.1 cm^3 of the diluted sample spread on agar gives 45 colonies. Estimate the number of bacteria per cm^3 of the original milk.

Solution: 45 colonies come from 0.1 cm^3 of diluted sample, so 1 cm^3 of diluted sample would give $45 \times 10 = 450$. Multiplying by the dilution factor gives $450 \times 1000 = 450{,}000 = 4.5 \times 10^5$ bacteria per cm^3 of original milk.

WORKED EXAMPLE 10.3

A mother with measles antibodies breast-feeds her baby, and the baby does not catch the disease for the first six months. Classify the immunity, and explain why the baby should still be vaccinated.

Solution: the baby received ready-made antibodies from its mother, so the immunity is natural passive. It does not involve the baby's own lymphocytes, so no memory cells form and it fades once the antibodies are broken down. Vaccination causes artificial active immunity with memory cells, so protection lasts for years.

WORKED EXAMPLE 10.4

Explain why the same drug used in a village outbreak could cure a bacterial infection such as typhoid but would be useless against measles.

Solution: antibiotics interfere with structures or processes found in bacteria, such as the cell wall or bacterial protein synthesis. Measles is caused by a virus, which has no cell wall or metabolism of its own and reproduces inside human cells, so antibiotics cannot affect it. Measles is prevented by vaccination.

10.9 Biotechnology and genetic engineering

Biotechnology is the use of living organisms or their products in industry, medicine and agriculture. Ancient forms include baking and brewing; modern forms include genetic engineering. In **genetic engineering** a gene from one organism is cut out using restriction enzymes, joined into a plasmid with ligase, and inserted into a host such as a bacterium. The bacterium then makes the protein coded by the foreign gene. Human insulin for diabetics is made this way. Genetically modified (GM) crops, such as insect-resistant cowpea, are intended to raise yields. Concerns include the effects on the environment, allergies and the cost of seeds. **Tissue culture**, growing plants from small pieces in nutrient jelly, is used to produce disease-free plantain, banana and yam in large numbers, and **cloning** produces genetically identical copies of an organism.

Exercise 10.1

1. State three differences between a virus and a bacterium.

2. Name the causative organism and the vector of (a) malaria; (b) sleeping sickness.
3. Explain how yeast makes bread rise and write the equation for the reaction involved.
4. Give three methods of preserving food and state how each prevents spoilage.
5. Distinguish between an epidemic, a pandemic and an endemic disease.
6. Describe how cholera is spread and how an outbreak may be controlled.
7. Distinguish between natural active and artificial passive immunity, with an example of each.
8. Explain why antibiotics are ineffective against viral diseases.
9. A bacterium divides every 30 minutes. How many bacteria will be present after 5 hours from a single cell?
10. A sample diluted 100 times gives 36 colonies from 0.2 cm^3 spread on agar. Estimate the number of bacteria per cm^3 in the undiluted sample.
11. Explain how a vaccine protects a person from a disease.
12. Name four ways in which the transmission of HIV can be prevented.

Answers to Exercise 10.1

1. Viruses are acellular but bacteria are cells; viruses reproduce only inside host cells, bacteria reproduce on their own by binary fission; viruses are much smaller; antibiotics kill bacteria but not viruses.

2. (a) *Plasmodium*, the female *Anopheles* mosquito; (b) *Trypanosoma*, the tsetse fly.
3. Yeast ferments sugar to ethanol and carbon dioxide; the gas forms bubbles that expand the dough. $C_6H_{12}O_6 \rightarrow 2C_2H_5OH + 2CO_2$.
4. Drying and salting remove water from microbes; refrigeration lowers the temperature and slows microbial growth; canning kills microbes by heat and seals out new ones.
5. An epidemic is a sudden outbreak in one area; a pandemic spreads across many countries; an endemic disease is constantly present in an area.
6. By drinking water or eating food contaminated with *Vibrio cholerae* from faeces, and by flies; control by safe water, sanitation, hand washing, oral rehydration and vaccination.
7. Natural active: recovery from chickenpox, where the person makes antibodies. Artificial passive: an injection of tetanus antiserum, giving ready-made antibodies for a short time.
8. Viruses lack a cell wall and their own metabolism, and live inside host cells, so antibiotics have no target in them.
9. 5 hours = 300 minutes; n = 300/30 = 10; $N = 2^{10} = 1{,}024$.
10. 0.2 cm³ gives 36, so 1 cm³ gives 180; × 100 = 18,000 = 1.8×10^4 per cm³.
11. It contains harmless antigens, which stimulate lymphocytes to make antibodies and memory cells; a later infection is then met by a rapid response.
12. Use of condoms, abstaining or fidelity, screening blood, using sterile needles, and treatment of infected mothers (any four).

Chapter summary

- Microorganisms include bacteria, fungi, protozoa and viruses; viruses are acellular and reproduce only inside host cells.

- Useful microbial activities include decomposition, nitrogen cycling, fermentation, antibiotic and enzyme production, and biogas; harmful ones include diseases, food spoilage and toxins.
- Food is preserved by drying, salting, smoking, cooling, canning and pasteurisation.
- Diseases spread by contact, droplets, contaminated food and water, vectors, body fluids and animals; important diseases include malaria, cholera, typhoid, tuberculosis, HIV/AIDS and Lassa fever.
- The body defends itself with barriers, phagocytes and antibody-producing lymphocytes; immunity may be active or passive, natural or artificial.
- Vaccines give active immunity; antibiotics kill bacteria but not viruses.
- Public health relies on clean water, sanitation, vector control, hygiene and immunisation.
- Bacterial numbers grow as $N = N_0 \times 2^n$; biotechnology includes genetic engineering, tissue culture and cloning.